

Session 2:

The Algorithmic Trading Process

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Session 1: Review



- Landscape
 - Electronic Trading, Algorithmic Trading
- Algorithms:
 - Execution-Algos, Quant-Trading, High Frequency Trading (HFT)
- Types of Algorithms:
 - Time Slicing (VWAP), Volume Follow (POV), Price Slicing (Arrival),
 - Traders Dilemma (Optimal) – Balance Tradeoff between MI & TR
- Execution Types:
 - Agency Execution, Principal Bid/Capital Commitment
- Trading Venues:
 - Exchange, Alternative Trading Systems (ATS), Dark Pools
- Smart Order Routing / Order Book:
 - Market Orders, Limit Orders
- Venue Revenue Model:
 - Commission, Maker-Taker, Inverted Maker-Taker

Session 2: Outline



- Transaction Costs
- Implementation Shortfall
- TCA: Post-Trade Analysis
 - Arrival Cost
 - VWAP Slippage
 - Relative Performance Measure
 - Value-Add
- TCA: Pre-Trade Analysis
 - Market Impact
 - Price Appreciation
 - Timing Risk
 - Evaluate Trading Strategy
 - Traders Dilemma - Optimization
- Python Scripts
 - Post-Trade Measure
 - Pre-Trade Analysis
 - Estimating MI Parameter

Which Investment would you Choose?



	<u>ABC</u>	<u>XYZ</u>
Return	10.0%	9.0%

Which Investment would you choose?



	<u>ABC</u>	<u>XYZ</u>
Return	10.0%	9.0%
Transaction Costs	<u>2.0%</u>	<u>0.5%</u>

Which Investment would you choose?



	<u>ABC</u>	<u>XYZ</u>
Return	10.0%	9.0%
Transaction Costs	<u>2.0%</u>	<u>0.5%</u>
Net Return	8.0%	8.5%

Which Investment would you choose?



	<u>ABC</u>	<u>XYZ</u>
Return	10.0%	9.0%
Transaction Costs	<u>2.0%</u>	<u>0.5%</u>
Net Return	8.0%	8.5%

Moral of the Story = Investors must choose their investments based on Net Return!

Important Benchmark Prices



Open = Opening price.

Close = Closing price.

VWAP = Volume weighted average price.

P_d = Decision price. Price at the time of the investment decision.

P_0 = Arrival price. Price at the time when the order was entered into the market.

P_n = Order end price. Price at the time when the order was completed or cancelled.

P_{Avg} = Average execution price. Average execution price of the order.

Transaction Costs

What are Transaction Costs?



- Total dollar amount paid above the decision price (premium) for buy orders and total dollars amount paid below the decision price (discount) for sell orders.
- Transaction costs are an unavoidable cost of business. They consist of nine (9) components: i) commissions, ii) spreads, iii) taxes, iv) fees & rebates, v) delay cost, vi) price appreciation, vii) market impact, viii) timing risk, and ix) opportunity cost.

Transaction Cost Iceberg



Depiction of the Iceberg to illustrate Transaction Costs was introduced by Wayne Wagner, The Plexus Group (1990).

Transaction Cost Components:

Visible Costs:

- **Commissions** = paid to brokers to execute the order.
- **Fees & Rebates** = payment to the trade venue (depending on the structure, some investors may pay a fee and other investors may receive a rebate).
- **Bid-Ask Spread** = difference between the ask price and bid price. It represents the total cost of trading, e.g., buy on the ask and sell on the bid.
- **Taxes** = SEC transfer tax paid by the seller.

Hidden Costs:

- **Delay Cost** = the cost due to the price change in the stock between the time of the investment decision and the time the order is entered into the market for execution.
- **Price Appreciation** = the cost due to buying stock in a rising market and/or selling stock in a falling market.
- **Market Impact** = the price movement in the stock due to the buying and/or selling pressure of the investor.
- **Timing Risk** = the cost due to market uncertainty, e.g., price volatility, volume variance, etc.
- **Opportunity Cost** = the missed profit opportunity for not being able to complete your entire order.

Implementation Shortfall: My Total Trading Cost.

This is the amount that I (the Portfolio Manager) must pay to execute my entire investment decision.

Implementation Shortfall is measured as the difference between the **Paper Return** of the portfolio and the **Actual Return** of the portfolio.

- **Paper Return** = My return if I were able to transact all shares of my order at my decision price and not incur any fixed costs such as broker commissions.
- **Actual Return** = My portfolio return based on execution prices and after accounting for all fixed fees such as broker commission.

Paper Return:

$$\text{Paper Return}(\$) = S \cdot (P_n - P_d)$$

Actual Return:

$$\text{Actual Return}(\$) = X \cdot (P_n - P_{avg}) - \text{Fixed Cost}$$

Implementation Shortfall:

$$IS(\$) = \text{Paper Return}(\$) - \text{Actual Return}(\$)$$

$$IS(\$) = S \cdot (P_n - P_d) - \{X \cdot (P_n - P_{avg}) - \text{Fixed Cost}\}$$

Definitions / Variables:

S = Order Shares

X = Executed Shares

P_d = Decision Price - mid-point of bid-ask spread at time of decision

P_{avg} = Average Execution Price of the Order

P_n = End Price - mid-point of bid-ask spread at end time of (completion/cancel)

Fixed Cost = Broker Commissions, Taxes, and Fees & Rebates

Spread Costs are embedded in the actual execution price of the stock

Fees & Rebates are often included in the broker commission

Implementation Shortfall: Example



Example:

A manager decides to purchase 100,000 shares of RLK at \$30.00. By the time the order is entered into the market the price has increased to \$30.15. The manager buys 80,000 shares at an average price of \$30.50. At the end of trading RLK is \$31.00. The broker commission cost for the trade is \$0.02/share.

Calculate the Implementation Shortfall of this Order:

$$\text{Paper Return}(\$) = 100,000 \cdot (\$31.00 - \$30.00) = \$100,000$$

$$\text{Actual Return}(\$) = 80,000 \cdot (\$31.00 - \$30.50) - 80,000 \cdot \$0.02 = \$40,000 - \$1,600 = \$38,400$$

$$\text{IS}(\$) = \text{Paper Return} - \text{Actual Return} = \$100,000 - \$38,400 = \$61,600$$

$$\text{IS}(\$) = \$61,600$$

It is important to understand where our trading costs occurring during implementation.

Implementation Shortfall can be decomposed into the following:

- **Delay Cost** = price movement in the stock from the time of the investment decision until the order is released to the market.
- **Execution Cost** = price movement in the stock over the time the order is being executed in the market.
- **Opportunity Cost** = the missed profit opportunity of not being able to execute all shares in the order.
- **Fixed Cost** = all fixed fees charged during implementation of the trade.

Implementation Shortfall Decomposition (Alternative Formula):

Formula:

$$IS_{\$} = \underbrace{S \cdot (P_0 - P_d)}_{\text{Delay Cost}} + \underbrace{X \cdot (P_{avg} - P_0)}_{\text{Execution Cost}} + \underbrace{R \cdot (P_n - P_0)}_{\text{Opportunity Cost}} + \text{Fixed Cost}$$

Variables:

S = Order Shares

X = Executed Shares

R = Unexecuted (Residual) Shares, $R = S - X$

P_d = Decision Price - mid-point of bid-ask spread at time of decision

P_0 = Arrival Price - mid-point of bid-ask spread at time of order entry

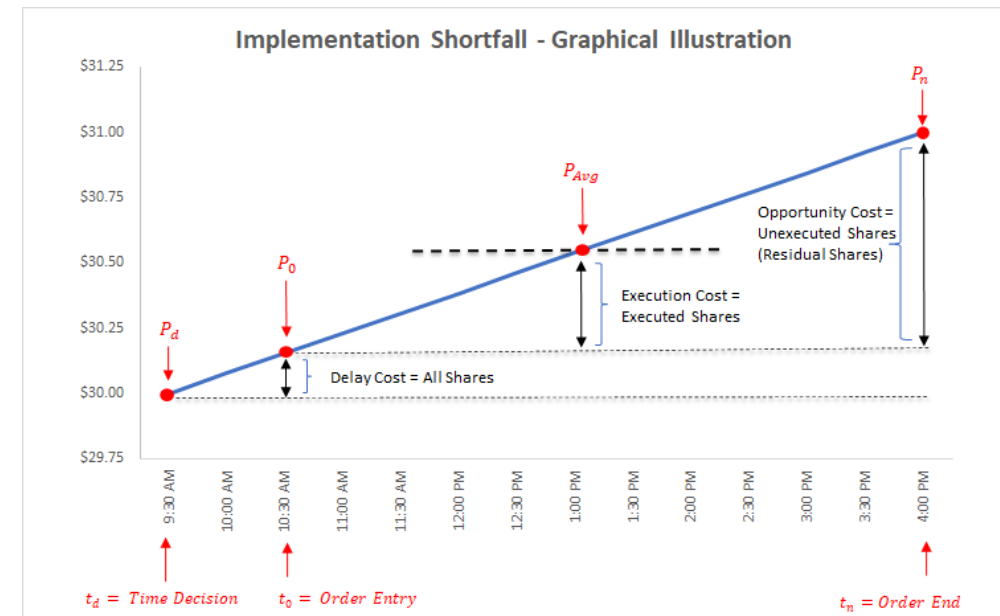
P_n = End Price - mid-point of bid-ask spread at end time of (completion/cancel)

P_{avg} = Average Execution Price of the Order

Fixed Cost = Commissions, Taxes, and Fees & Rebates

Spread Costs are embedded in the actual execution price of the stock

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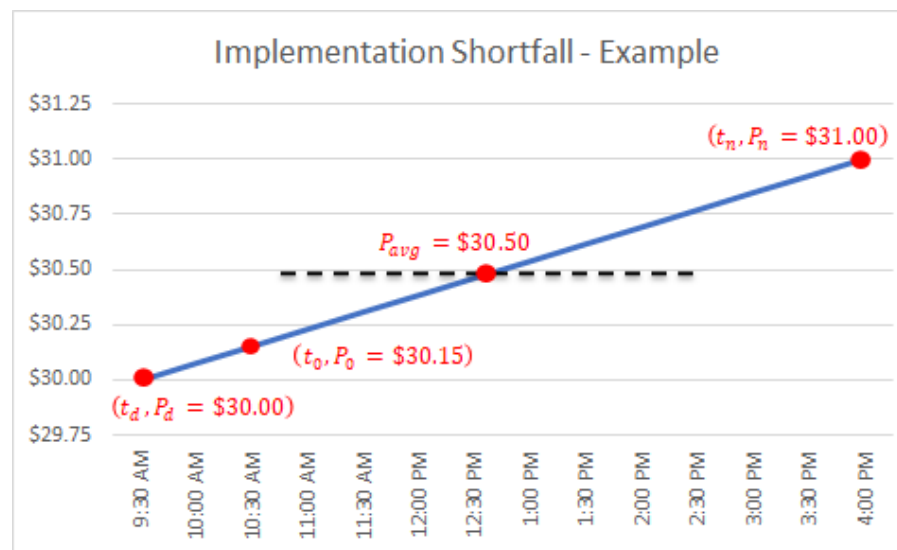
Implementation Shortfall: Example Revisited

Example:

A manager decides to purchase 100,000 shares of RLK at \$30.00. By the time the order is entered into the market the price has increased to \$30.15. The manager buys 80,000 shares at an average price of \$30.50. At the end of trading RLK is \$31.00. The broker commission cost for the trade is \$0.02/share.

Calculate the Implementation Shortfall of this trade and all cost components.

Implementation Shortfall - Example						
Pd=	\$30.00	Order Shares =	100,000			
P0=	\$30.15	Executed =	80,000			
Pn=	\$31.00	Residual =	20,000			
Pavg=	\$30.50					
comm =	\$0.02	Order Value=	\$3,000,000			
Component		Shares	Chg	Dollars	\$/Share	bp
Delay	Pd-P0	100,000	\$0.15	\$15,000	\$0.15	50.0
Execution	Pavg-P0	80,000	\$0.35	\$28,000	\$0.28	93.3
Opportunity	Pn-P0	20,000	\$0.85	\$17,000	\$0.17	56.7
Fixed	comm.	80,000	\$0.02	\$1,600	\$0.02	5.3
Total				\$61,600	\$0.62	205.3



$$IS_{\$} = \underbrace{100,000 \cdot (\$30.15 - \$30.00)}_{\text{Delay} = \$15,000} + \underbrace{80,000 \cdot (\$30.50 - \$30.15)}_{\text{Execution} = \$28,000} + \underbrace{20,000 \cdot (\$31.00 - \$30.15)}_{\text{OC} = \$17,000} + \underbrace{80,000 \cdot \$0.02}_{\text{Fixed} = \$1,600} = \$61,600$$

Implementation Shortfall (IS)

Historical (Past):

$$IS = \underbrace{\left\langle \begin{matrix} Delay \\ Cost \end{matrix} \right\rangle}_{\text{Before}} + \underbrace{\left\langle \begin{matrix} Execution \\ Cost \end{matrix} \right\rangle}_{\text{During}} + \underbrace{\left\langle \begin{matrix} Opportunity \\ Cost \end{matrix} \right\rangle}_{\text{After}} + \underbrace{\left\langle \begin{matrix} Fixed \\ Costs \end{matrix} \right\rangle}_{\text{Visible}}$$

Post Trade Analysis

Trade Execution Performance:

- **Transaction Cost Analysis (TCA) / Post Trade Analysis / Post Trade Measurement**
- How do we determine if our broker/algorithm did a good job?
- We need to evaluate the Arrival Cost (Trade Cost/Execution Cost) component.
- This is the cost that arises during trading and is due to the broker/algorithm.
- $\text{Arrival Cost} = \text{Execution Cost Component of the Implementation Shortfall}$.

Performance Measures:

- **Arrival Cost** = The cost of the order compared to the market price at the time of order entry. This is the cost attributable to the broker/algorithm.
- **VWAP Slippage** = A comparison of your execution price to the market VWAP price. The market VWAP price is a proxy for the average execution price over the trading horizon.
- **RPM** = Relative Performance Measure. A percentile ranking of your trading performance. For example, RPM=90% indicates your trading was in the top 90% of all executions.
- **Value-Add** = A measure of how well the broker/algorithm performed compared to the expected cost of the trade based on actual market conditions.

Arrival Cost Calculation



Arrival Cost = what was my cost incurred during implementation of the order?

Arrival cost is most commonly expressed in basis points.

$$Arrival\ Cost(bp) = Side \cdot \frac{P_{avg} - P_0}{P_0} \cdot 10^4_{bp}$$

where,

P_{avg} = Average Execution Price of the order

P_0 = Arrival Price of the order (when the order was released into the market)

$Side$ = 1 if Buy, -1 if Sell

Basis Points:
Decimal = 0.03

Percent = 3% (0.03 x 100 = 3%)

Basis Point = 300
3% x 100 = 300
0.03 x 100 x 100 = 300
0.03 x 10,000 = 300

Cost = 0.00025
Cost = 0.025%
Cost = 2.5 bp

Execution Cost / Arrival Cost / Trade Cost = these terms are exactly the same!

VWAP Slippage = how did I do compared to all other participants?

VWAP slippage is expressed in basis points.

$$VWAP\ Slippage(bp) = Side \cdot \frac{P_{avg} - VWAP}{VWAP} \cdot 10^4_{bp}$$

where,

P_{avg} = Average Execution Price of the order

$VWAP$ = Market VWAP Price (over the trading horizon)

$Side$ = 1 if Buy, -1 if Sell

TCA: Benchmark Analysis (most often calculated in basis points)



Example:

A portfolio manager bought 100,000 shares of a 150,000 shares order at an average price of $P_{avg} = \$50.75$. The benchmark prices on the day are as follows: Open = \$50.15, Arrival = \$50.30, VWAP = \$50.72, and Close = \$50.90. Calculate the Arrival Cost and VWAP Slippage.

Arrival Cost:

$$\text{Arrival Cost (bp): } +1 \cdot \frac{\$50.75 - \$50.30}{\$50.30} \cdot 10^4 = 89.46 \text{ bp}$$

VWAP Slippage:

$$\text{VWAP Slippage (bp): } +1 \cdot \frac{\$50.75 - \$50.72}{\$50.72} \cdot 10^4 = 5.91 \text{ bp}$$

Relative Performance Measure (RPM)

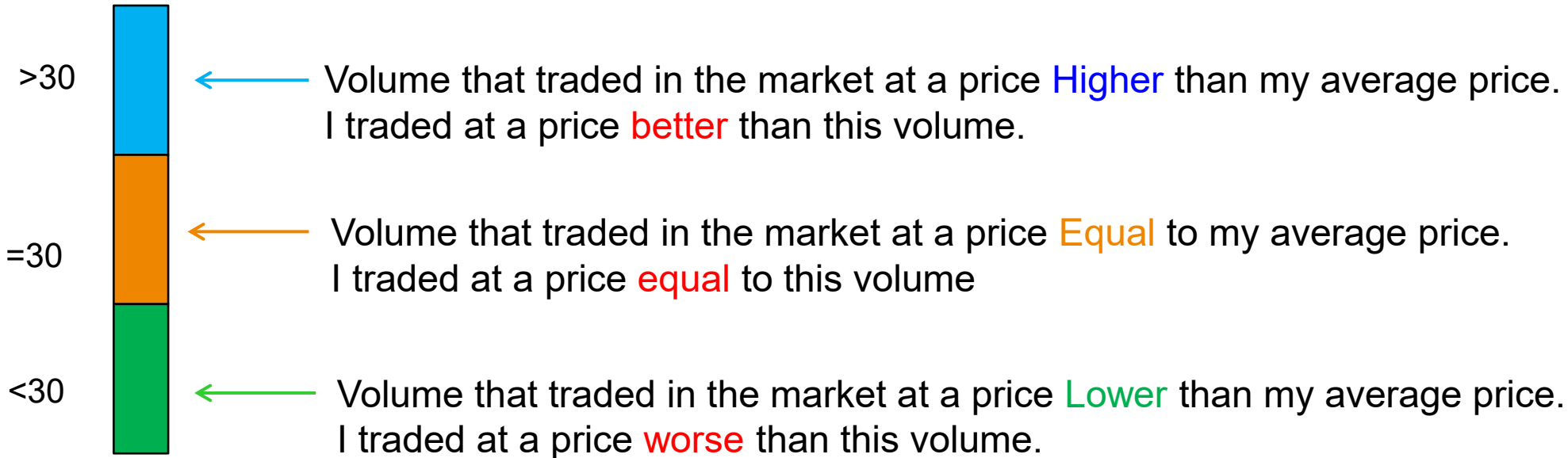


RPM: The Relative Performance Measure (RPM) is a percentile ranking of your trading activity. It provides the percentage of trading activity that the investor outperformed in the market.

- For a buy order, it represents the percentage of market activity that transacted at a higher price.
- For a sell order, it represents the percentage of market activity that transacted at a lower price.
- The RPM is modeled after the percentile ranking used in standardized academic tests and provides a descriptive statistic that is more consistent and robust than other measures.

Relative Performance Measure (RPM) – Buy Order

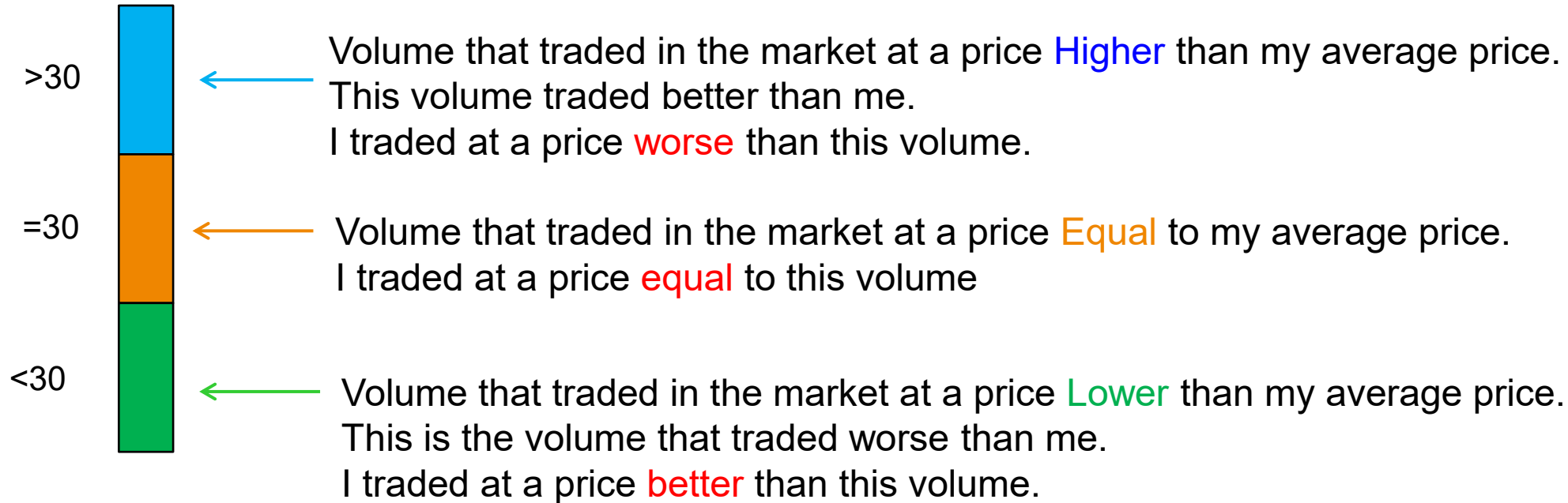
I traded my Buy Order at an average Price of \$30.00



$$\text{RPM}(\text{Buy}) = \frac{\text{Volume at Price} > P_{Avg} + 0.5 \cdot \text{Volume at } P_{Avg}}{\text{Total Volume}}$$

Relative Performance Measure (RPM) – Sell Order

I traded my Sell Order at an average Price of **\$30.00**



$$\text{RPM}(\text{Sell}) = \frac{\text{Volume at Price} < P_{Avg} + 0.5 \cdot \text{Volume at } P_{Avg}}{\text{Total Volume}}$$

Relative Performance Measure - Example



Example:

The execution price for an order was $P_{avg} = \$25.00$. The total market volume was 100,000 shares. The volume greater than \$25.00 was 20,000 shares. The volume less than \$25.00 was 50,000 shares.

- What is the RPM if this was a Buy Order?

Information:

Total Volume = 100,000

Volume worse (higher) than My Buy Order Price = 20,000 Shares

Volume better (lower) than My Buy Order Price = 50,000 Shares

Volume equal to My Buy Order Price = 30,000 Shares

$$RPM(Buy) = \frac{20,000 + 0.5 \cdot 30,000}{100,000} = \frac{35,000}{100,000} = 35\%$$

Relative Performance Measure - Example



Example:

The execution price for an order was $P_{avg} = \$25.00$. The total market volume was 100,000 shares. The volume greater than \$25.00 was 20,000 shares. The volume less than \$25.00 was 50,000 shares.

- What is the RPM if this was a Sell Order?

Information:

Total Volume = 100,000

Volume worse (lower) than My Sell Order Price = 50,000 Shares

Volume better (higher) than My Sell Order Price = 20,000 Shares

Volume equal to My Sell Order Price = 30,000 Shares

$$RPM(Sell) = \frac{50,000 + 0.5 \cdot 30,000}{100,000} = \frac{65,000}{100,000} = 65\%$$

Value-Add Analysis is used to determine if the realized transaction costs were appropriate given the order characteristics (size) actual market conditions (volatility and volume) during the trading horizon.

A **Z-Score** is used to adjust the value-add metric for the risk of the trade. Risk-Adjusted Value-Add metric.

These value-add metrics are calculated using the Arrival Cost of the trade and the Expected Cost for the actual number of shares executed in the market under actual market conditions.

- Expected Cost is calculated as the expected market impact cost for the number of shares executed.
- Expected Market Impact cost is calculated based on actual market conditions.
- We can also make an adjustment for general market movement, e.g., Beta Adjusted Cost.

Depending on the needs of the investor, the Estimated Cost may include an adjustment for general market movement, such as a Beta Adjusted Cost.

- A positive Value-Add indicates out-performance.
- A negative Value-Add metric indicates under-performance.

These formulas are as follows:

$$\text{Value Add} = \text{Estimated Cost} - \text{Arrival Cost}$$

$$\text{Z Score} = \frac{\text{Estimated Cost} - \text{Arrival Cost}}{\text{Timing Risk}} = \frac{\text{Value Add}}{\text{Timing Risk}}$$

Example:

An investor executed 100,000 shares at an Arrival Cost of 60bp.

The Estimated Cost of the order was 50bp and Timing Risk of the order was 90bp.

Calculate the Broker Value-Add and Z-Score.

Solution:

$$\text{Value Add} = 50bp - 60bp = -10bp$$

$$\text{Z Score} = \frac{50 - 60}{90} = \frac{-10}{90} = -0.1111$$

Pre-Trade Analysis

Historical compared to Future

Historical (Past):

$$IS = \underbrace{\left\langle \begin{array}{c} \textit{Delay} \\ \textit{Costs} \end{array} \right\rangle}_{\text{Before}} + \underbrace{\left\langle \begin{array}{c} \textit{Execution} \\ \textit{Costs} \end{array} \right\rangle}_{\text{During}} + \underbrace{\left\langle \begin{array}{c} \textit{Opportunity} \\ \textit{Cost} \end{array} \right\rangle}_{\text{After}} + \underbrace{\left\langle \begin{array}{c} \textit{Fixed} \\ \textit{Costs} \end{array} \right\rangle}_{\text{Visible}}$$

Forecast (Future):

$$\left\langle \begin{array}{c} \textit{Execution} \\ \textit{Costs} \end{array} \right\rangle = \textit{Expected} \underbrace{\left\langle \begin{array}{c} \textit{Market Impact (MI)} \\ \textit{Price Appreciation (PA)} \\ \textit{Timing Risk (TR)} \end{array} \right\rangle}_{\text{Estimate/Forecast}}$$

Past (Post-Trade)

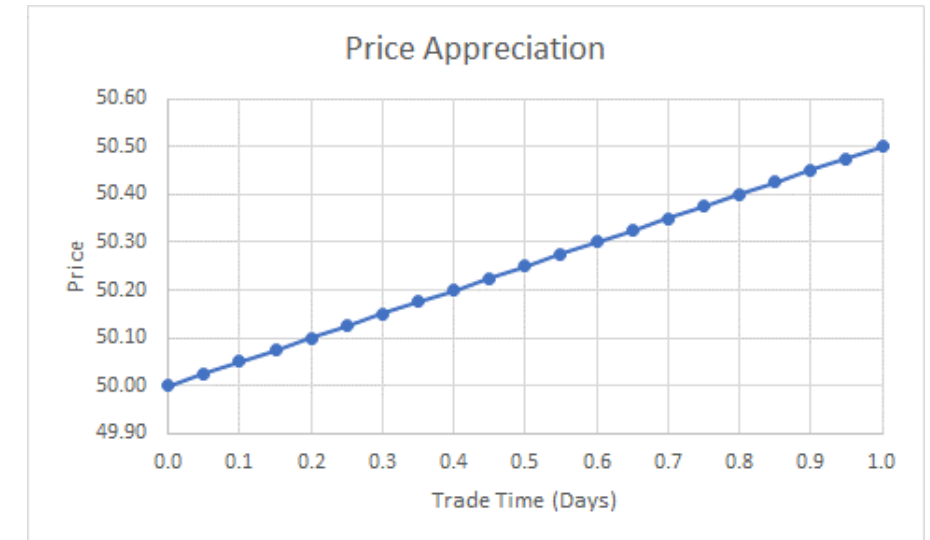
- Cost Measure
 - Single Value, Cost is Known (Past)
 - Implementation Shortfall
 - Delay, Execution, Opportunity, Fixed Cost
 - Benchmark TCA:
 - Arrival, VWAP, RPM, Value-Add
- $$\text{Arrival Cost} = \left(\frac{P_{avg} - P_0}{P_0} \right) \cdot 10^4 bp$$

Forward (Pre-Trade)

- Forecast/Estimation
 - Market Impact: MI
 - Price Appreciation: PA
 - Timing Risk: TR
- Distribution of Costs
 - PDF and CDF Graphs
 - Mean = MI + PA
 - Stdev = TR
- Optimization
 - Optimal Trading Rates

Price Appreciation (PA)

- Price Appreciation (PA) is the natural price movement of the stock. It represents how stock price would evolve in a market without any uncertainty.
- Price appreciation is also referred to as price trend, momentum, drift, or alpha.
- Buy Order = It represents a cost when buying stock in a rising market and a savings when buying stock in a falling market.
- Sell Order = It represents a cost when selling stock in a falling market and a savings when selling stock in a rising market.
- AlphaBp = the estimated price appreciation over one-day expressed in basis points.
- Side = +1 for Buy order, -1 for a Sell order.
- Size = Shares divided by Stock Average Daily Volume (ADV)



Time:

$$PA = Side \cdot \frac{1}{2} \cdot \text{AlphaBp} \cdot \text{Trade Time}$$

POV Rate:

$$PA = Side \cdot \frac{1}{2} \cdot \text{AlphaBp} \cdot \frac{\text{Shares}}{\text{ADV}} \cdot \frac{(1 - POV)}{POV}$$

Time-POV Relationship:

$$\text{Trade Time} = \frac{\text{Shares}}{\text{ADV}} \cdot \frac{(1 - POV)}{POV} = \text{Size} \cdot \frac{(1 - POV)}{POV}$$

$$POV = \frac{\text{Size}}{\text{Size} + \text{Trade Time}}$$

$$\text{Size} = \frac{\text{Shares}}{\text{ADV}}$$

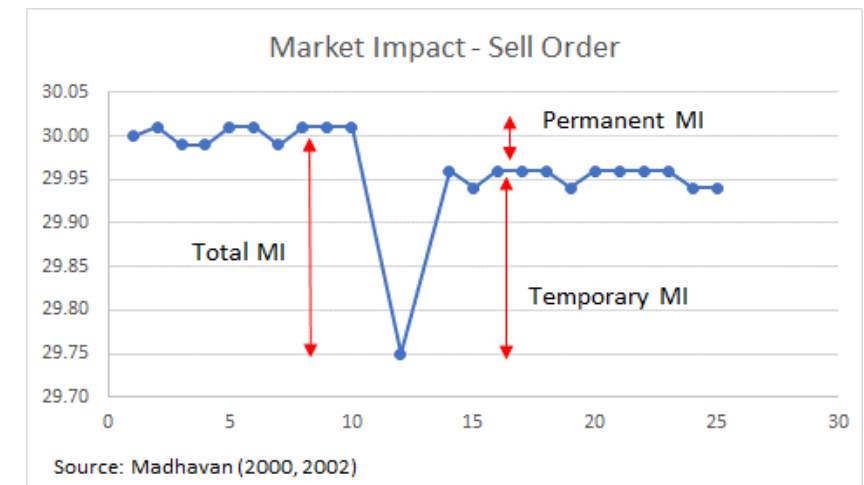
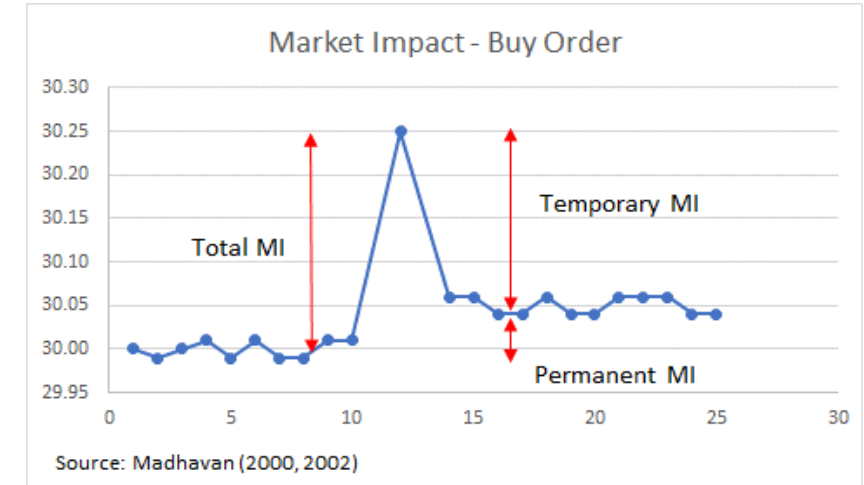
Market Impact (MI)

- Market Impact (MI) is the price movement in a stock caused by the order or trade. It consists of two components:
 - Temporary Impact = due to the liquidity needs of the investor.
 - Permanent Impact = due to the information content of the trade.
- Mathematically, MI is defined as the difference between the path of the stock with and without the order.
- Heisenberg Uncertainty Principal of Trading.
- Linear Regression Model: $y = b_0 + b_1 \cdot x_1 + b_2 \cdot x_2 + e$
- $Size = \frac{Shares}{ADV}$ \$100,000: 1bp = 0.0001 x \$100,000 = \$10

Market Impact (MI)

$$I_{bp}^* = a_1 \cdot Size^{a_2} \cdot \sigma^{a_3}$$

$$MI_{bp} = I_{bp}^* \cdot (b_1 \cdot POV^{a_4} + (1 - b_1))$$

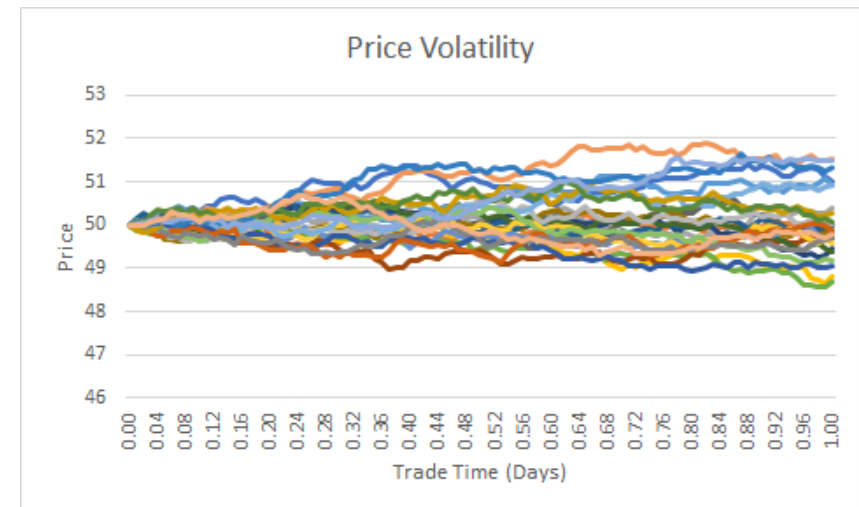
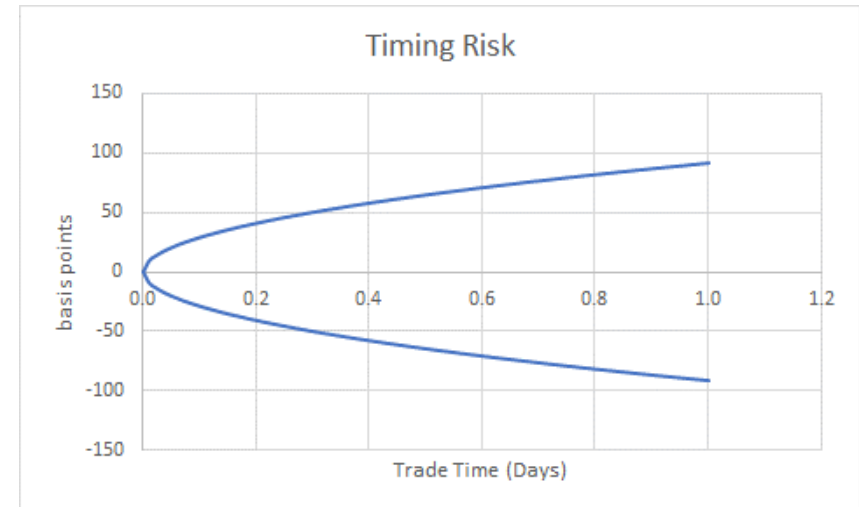


Timing Risk (TR)

- Timing Risk (TR) refers to the uncertainty surrounding the estimated trading cost. Timing risk can result in the execution price being higher or lower than the arrival price. Timing Risk is the Standard Error of our Cost Estimate.
- It consists of three components
 - ❖ Price Volatility
 - ❖ Volume Uncertainty (Liquidity Risk)
 - ❖ Parameter Estimation Error

Timing Risk (TR)

$$TR_{bp} = \sigma \cdot \sqrt{\frac{1}{250} \cdot \frac{1}{3} \cdot \frac{Shares}{ADV} \cdot \frac{1 - POV}{POV}} \cdot 10^4_{bp}$$



Trade Cost Equations

Market Impact (MI):

$$I_{bp}^* = a_1 \cdot Size^{a_2} \cdot \sigma^{a_3}$$



My Effect on Stock Price

$$MI_{bp} = I_{bp}^* \cdot (b_1 \cdot POV^{a_4} + (1 - b_1))$$

$$\rightarrow MI_{bp} = (a_1 \cdot Size^{a_2} \cdot \sigma^{a_3}) \cdot (b_1 \cdot POV^{a_4} + (1 - b_1))$$

Timing Risk (TR):

$$TR_{bp} = \sigma \cdot \sqrt{\frac{1}{250} \cdot \frac{1}{3} \cdot Size \cdot \frac{1 - POV}{POV}} \cdot 10^4_{bp}$$



Stock Price Movement from everyone else and noise and randomness

Price Appreciation (PA):

$$PA_{bp} = Side \cdot \frac{1}{2} \cdot AlphaBp \cdot Size \cdot \frac{1 - POV}{POV}$$



Stock Price Movement by Itself

Trade Cost Equations

Market Impact (MI):

$$I_{bp}^* = a_1 \cdot Size^{a_2} \cdot \sigma^{a_3}$$

$$MI_{bp} = (b_1 \cdot I_{bp}^* \cdot POV^{a_4} + (1 - b_1) \cdot I_{bp}^*)$$

Timing Risk (TR):

$$TR_{bp} = \sigma \cdot \sqrt{\frac{1}{250} \cdot \frac{1}{3} \cdot Size \cdot \frac{1 - POV}{POV}} \cdot 10^4_{bp}$$

Price Appreciation (PA):

$$PA_{bp} = Side \cdot \frac{1}{2} \cdot AlphaBp \cdot Size \cdot \frac{1 - POV}{POV}$$

Model Parameters:

- a_1, a_2, a_3, a_4, b_1

Variables:

- Size
- Volatility
- Side
- AlphaBp

Decision Variables:

- POV Rate
- Trade Time

Example: Pre-Trade Analysis



Example:

An investor has a buy order for of 10% ADV for stock RLK. RLK has volatility of 35% and the investor estimates price appreciation of **AlphaBp=100bp**. If the order is going to be executed using a POV=15% strategy. Calculate the MI, TR, and PA of the order.

The market impact parameters are: **a1 = 883.59, a2 = 0.35, a3 = 0.76, a4 = 0.83, b1 = 0.96**

Market Impact:

$$MI = (a_1 \cdot Size^{a_2} \cdot \sigma^{a_3}) \cdot (b_1 \cdot POV^{a_4} + (1 - b_1))$$

$$MI = (883.59 \cdot 0.10^{0.35} \cdot 0.35^{0.76}) \cdot (0.96 \cdot 0.15^{0.83} + (1 - 0.96)) = 41.99bp$$

Timing Risk:

$$TR_{bp} = \sigma \cdot \sqrt{\frac{1}{250} \cdot \frac{1}{3} \cdot Size \cdot \frac{1 - POV}{POV}} \cdot 10^4_{bp} = 0.35 \cdot \sqrt{\frac{1}{250} \cdot \frac{1}{3} \cdot 0.10 \cdot \frac{1 - 0.15}{0.15}} \cdot 10^4_{bp} = 96.2$$

Price Appreciation:

$$PA_{bp} = Side \cdot \frac{1}{2} \cdot AlphaBp \cdot Size \cdot \frac{1 - POV}{POV} = +1 \cdot \frac{1}{2} \cdot 100 \cdot 0.10 \cdot \frac{1 - 0.15}{0.15} = 28.3$$

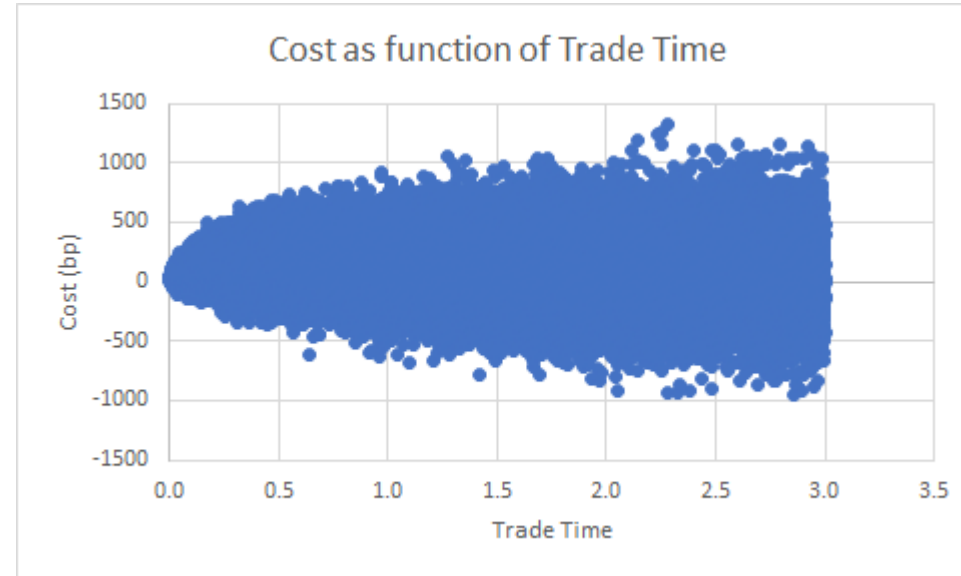
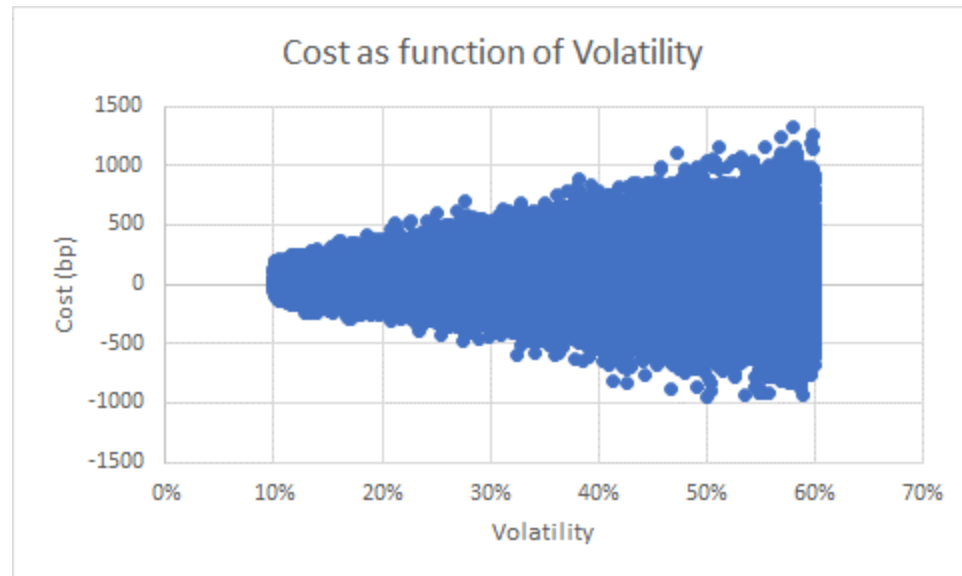
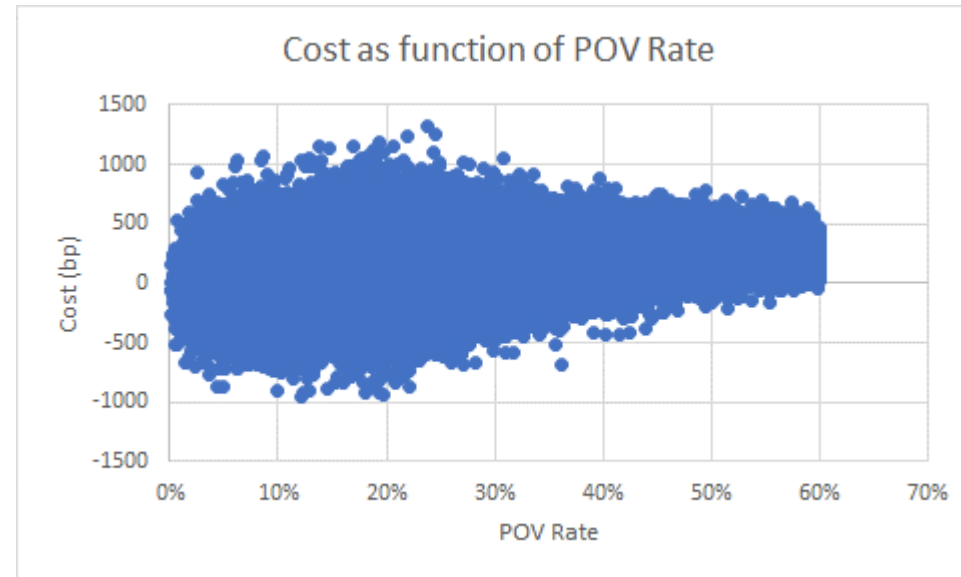
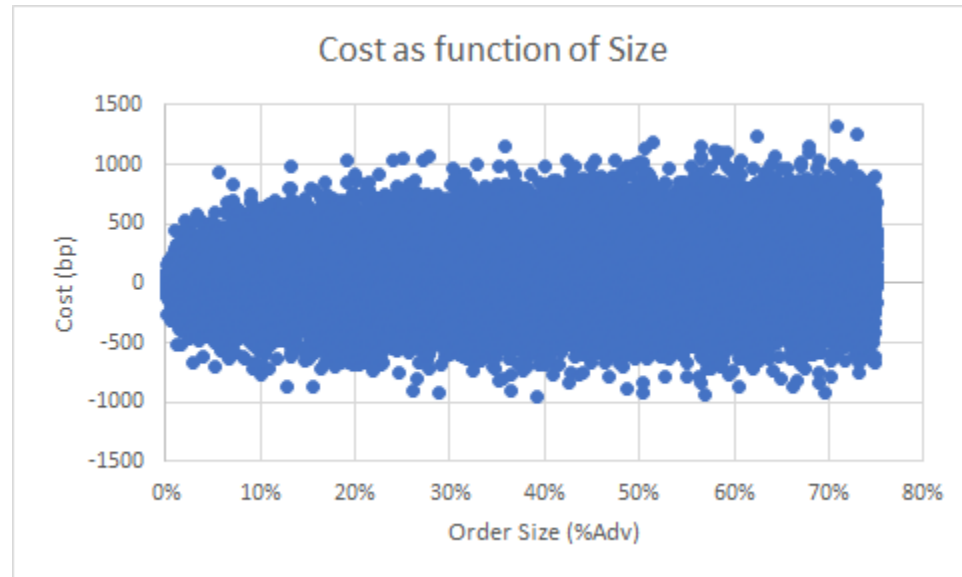
Estimating MI Parameters

What MI Explanatory Factors are Important?



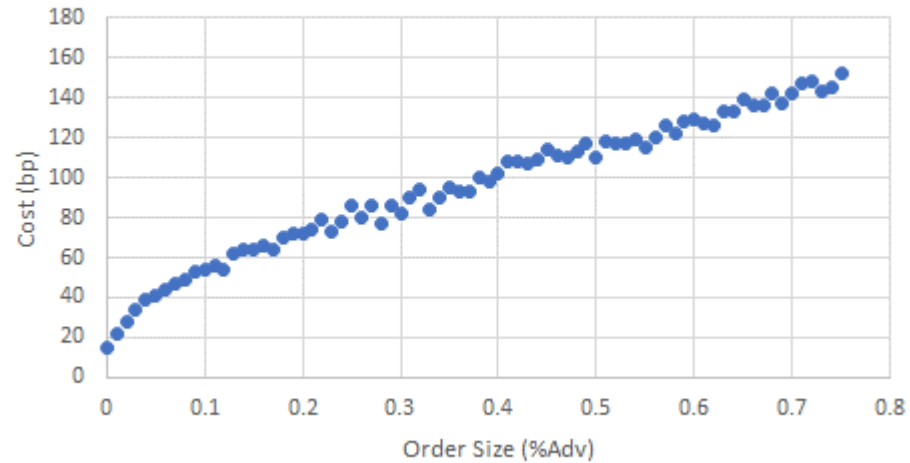
- Share Quantity
 - Size (%Adv)
- Volatility
 - Annualized
- POV / Trading Rate
 - Strategy Component

Historical Trade Cost Data (Individual Observations)

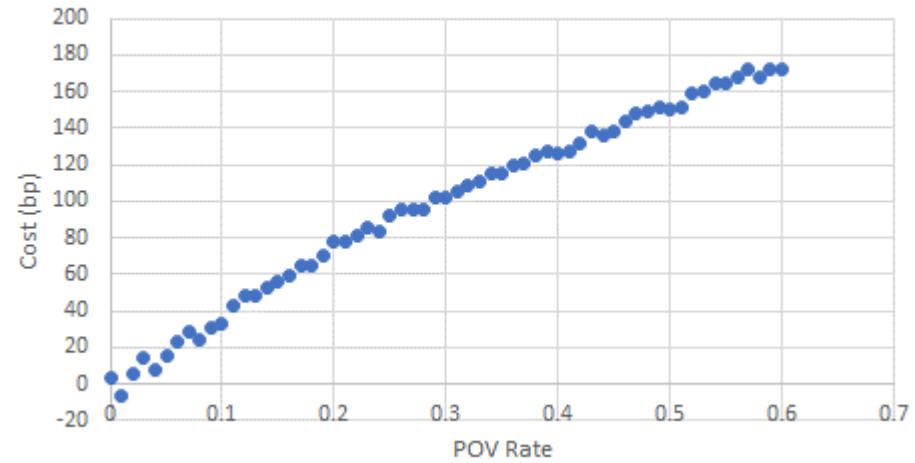


Historical Trade Cost Data (Grouped Data)

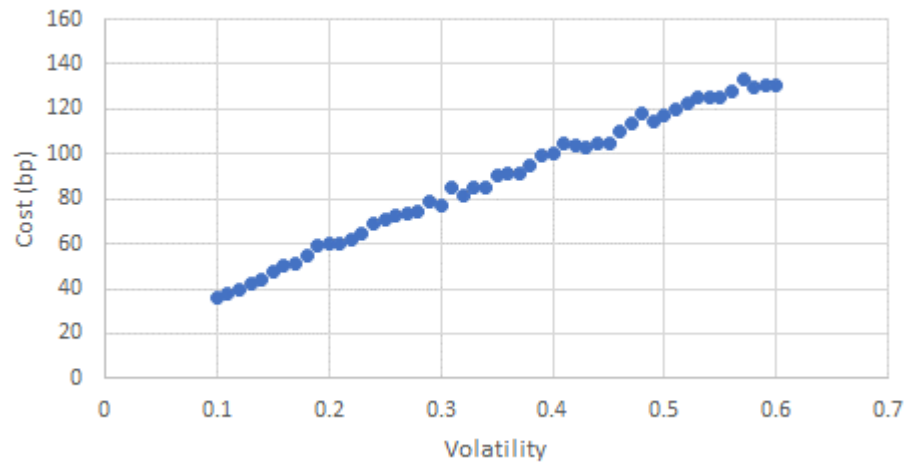
Cost as function of Order Size



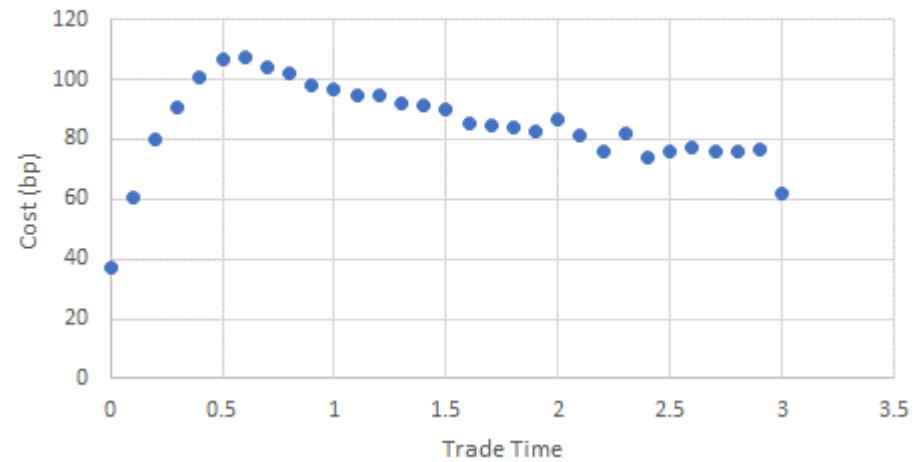
Cost as function of POV Rate



Cost as function of Volatility



Cost as function of Trade Time



Estimating MI Parameters: Non-Linear Regression



I-Star Model:

$$MI = (a_1 \cdot Size^{a_2} \cdot \sigma^{a_3}) \cdot (b_1 \cdot POV^{a_4} + (1 - b_1))$$

Non-Linear Regression: Minimize the SSE (sum of the squared errors)

$$\text{Min } f = \sum (Actual\ Arrival\ Cost - MI)^2$$

$$\text{Min } f = \sum \left(Actual\ Arrival\ Cost - \left((a_1 \cdot Size^{a_2} \cdot \sigma^{a_3}) \cdot (b_1 \cdot POV^{a_4} + (1 - b_1)) \right) \right)^2$$

Subject to:

$$0 \leq b_1 \leq 1$$

$$a_1, a_2, a_3, a_4 \geq 0$$

where,

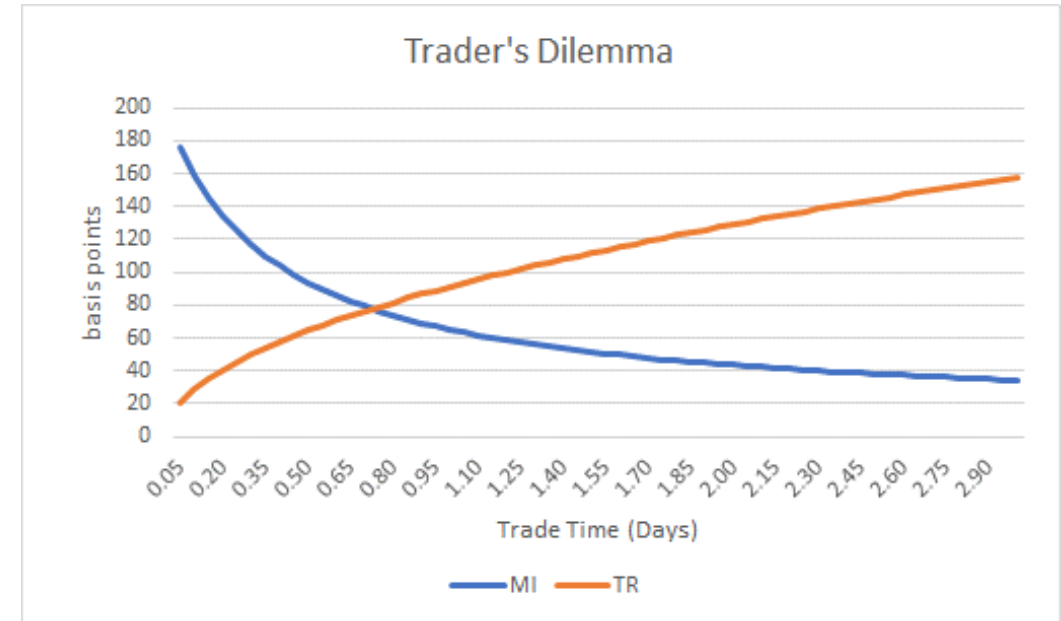
Actual Arrival Cost = Measured Arrival Cost (difference between execution price and arrival price)

Traders Dilemma

Traders Dilemma

Traders Dilemma:

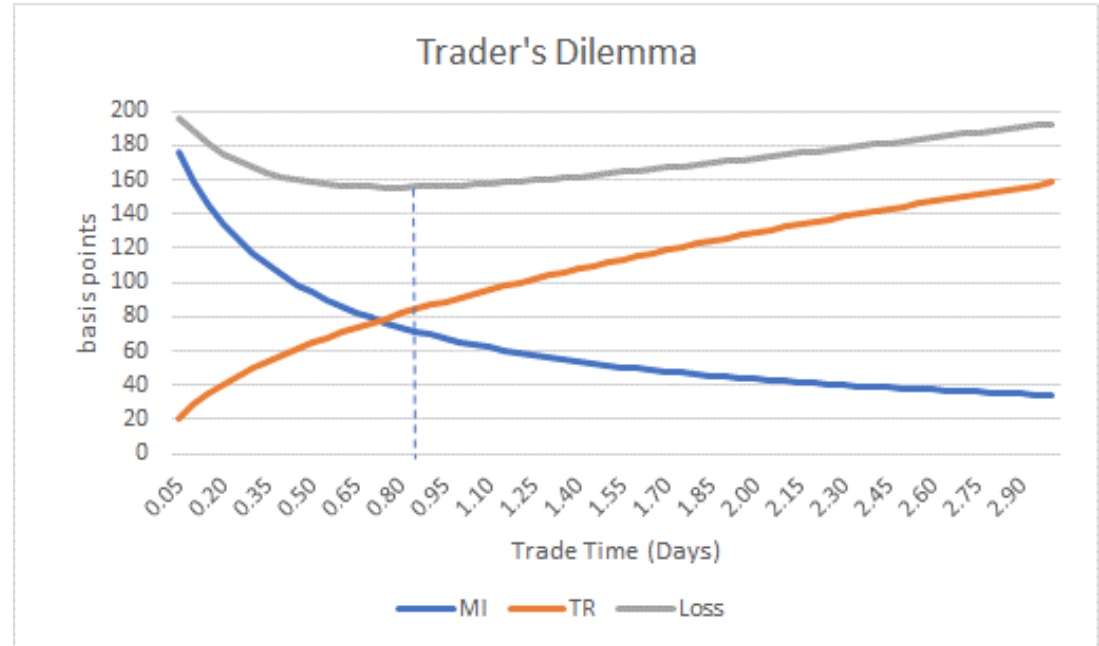
- *Trade too fast and I will move the market. Trade too slow and the market will move me.*
- *Market Impact and Timing Risk are conflicting expressions.*



Solving Trader Dilemma

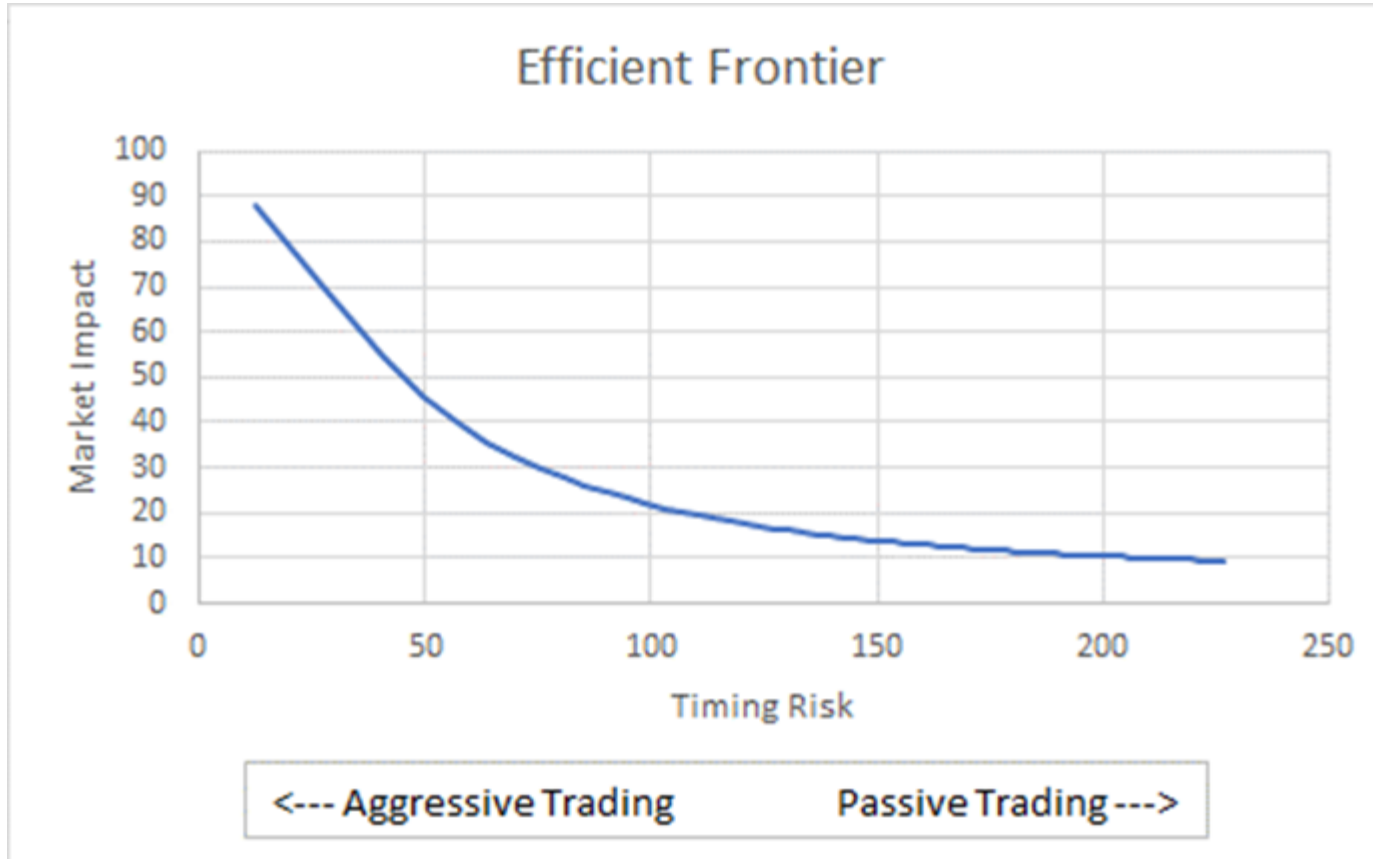
Solving Traders Dilemma:

- Traders seek to manage the tradeoff between market impact cost (MI) and timing risk (TR) at a specified level of risk aversion λ .
- Lambda λ is specified by the investor and is often $0 \leq \lambda \leq 3$.
- Solve via Optimization
 - Solution provides optimal POV rate
- Each value of Lambda will have a different “Optimal” solution.



$$\text{Min} \quad MI + \lambda \cdot TR$$

Efficient Trading Frontier



Optimization Results				
<u>Lambda</u>	<u>POV</u>	<u>Time</u>	<u>MI</u>	<u>TR</u>
0.10	7%	1.25	14.6	142.9
0.25	14%	0.60	21.9	99.0
0.35	18%	0.45	25.9	85.7
0.50	25%	0.30	32.4	70.0
0.75	33%	0.20	40.1	57.2
1.00	50%	0.10	54.6	40.4
2.00	90%	0.01	87.7	12.8

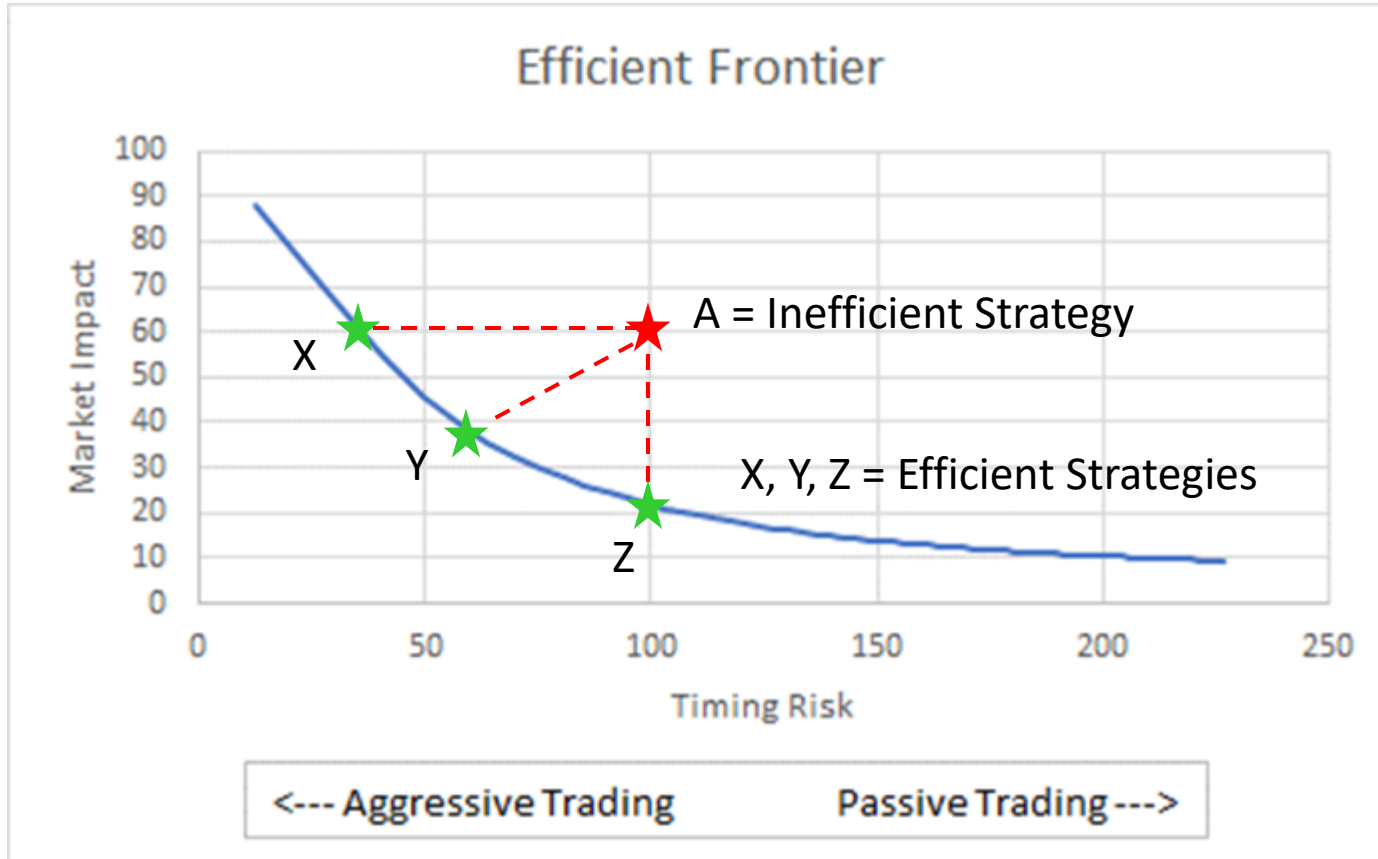
$$\text{Min} \quad MI + \lambda \cdot TR$$

Efficient Trading Frontier: Almgren & Chriss (1997)

Source: Kissell Research Group (2022)

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Efficient Trading Frontier

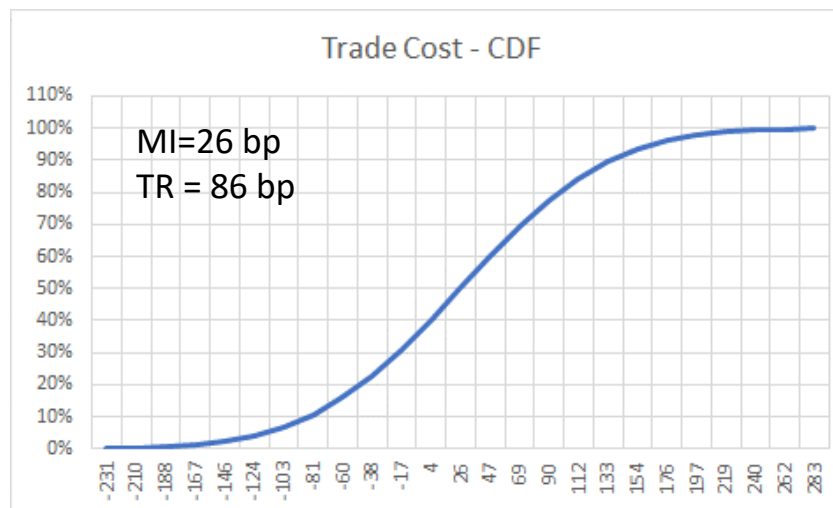
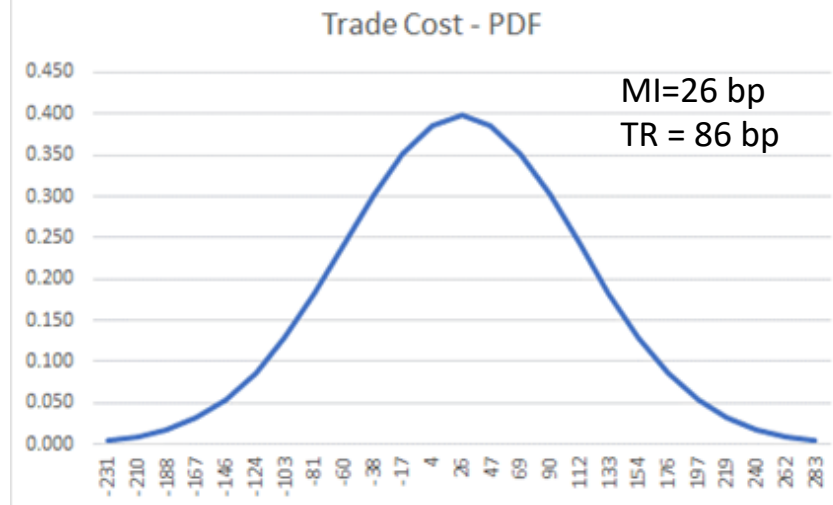
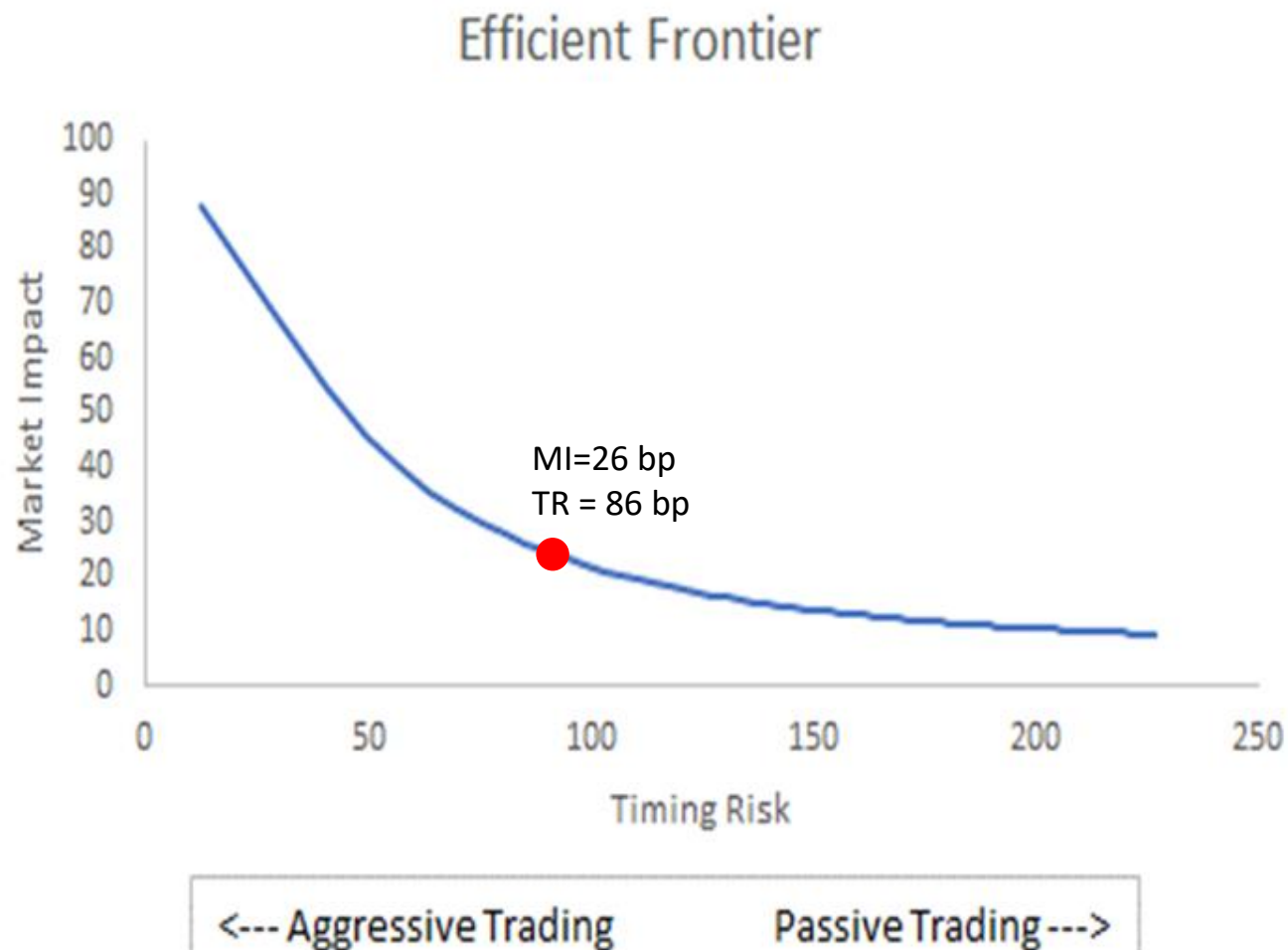


Optimization Results				
Lambda	POV	Time	MI	TR
0.10	7%	1.25	14.6	142.9
0.25	14%	0.60	21.9	99.0
0.35	18%	0.45	25.9	85.7
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2.00	90%	0.01	87.7	12.8

Source: Kissell Research Group (2022)

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Where are the Expected Costs?



Source: Kissell Research Group (2022)

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Python Examples

Python TCA Scripts:

- [KRG-Est_MI-Parameters.ipynb](#)
- [KRG-Calc_Post-Trade.ipynb](#)
- [KRG-Calc_Pre-Trade.ipynb](#)

Appendix

Questions

Q1) A trader received an order with the following characteristics: $Size = 15\%$, $\sigma = 25\%$, and $POV = 20\%$. What is the MI & TR of this order?

Q2) A trader received a buy order for 75,000 shares of stock RLK with instructions to transact the order over one-full day, e.g., 1-day VWAP Strategy. The statistics for RLK are: $ADV = 500,000$ and $\sigma = 25\%$. What is the MI & TR of this order?

Q3) A trader received an order to sell 100,000 shares of RLK using a $POV=15\%$ strategy. The stock characteristics are: $ADV = 1,000,000$ and $\sigma = 20\%$.

- How long will it take to trade this order (e.g., Time=Volume Time)?
- What is the MI & TR of this order?

Q4) A trader received an order with the following characteristics: $Size = 15\%$, and $\sigma = 25\%$. The trader has a risk aversion of $\lambda = 0.35$. What is the optimal POV rate that balances the tradeoff of MI & TR of this order at the specified level of risk aversion?

Q5) A trader will execute the order using a $POV=20\%$ strategy during a period where the expected period volume is 50,000 shares. How The stock has $ADV=1,000,000$ shares. How many shares will the trader during this period?

Parameter	Value
a1	883.25
a2	0.35
a3	0.76
a4	0.83
b1	0.96

Trade Cost Equations:

$$I^*(bp) = a_1 \cdot Size^{a_2} \cdot \sigma^{a_3}$$

$$MI(bp) = b_1 \cdot I^* \cdot POV^{a_4} + (1 - b_1) \cdot I^*$$

$$TR(bp) = \sigma \cdot \sqrt{\frac{1}{250} \cdot \frac{1}{3} \cdot Size \cdot \frac{1 - POV}{POV}} \cdot 10^4_{bp}$$

$$Time = Size \cdot \frac{1 - POV}{POV}$$

Spread Cost

- **Spread** = difference between best offer (ask) and best bid price.
 - ❖ Represent the round-trip cost of transacting small orders, e.g., buying and immediately selling 100 shares.
 - ❖ Does not accurately represent the cost of transacting larger share quantities and blocks.
- **Spread Cost** = **one-half of the Bid-Ask Spread**. Measured as the midpoint of the bid-ask spread to the market price. The spread cost for buy and sell orders are:
 - ❖ **Buy Order = Ask Price – Mid Point of Bid Ask Spread**
 - ❖ **Sell Order = Mid Point of Bid Ask Spread – Bid Price**
- Spread Cost is often incorporated into the Market Impact Cost estimate, i.e., arrival price P_0 is taken as the mid-point of the bid-ask spread.

Order Book

	Time	Shares	Price
Sell Orders (Asks)	10:00 AM	500	30.20
	10:20 AM	1000	30.15
	10:10 AM	100	30.15
	10:00 AM	500	30.15
	10:05 AM	200	30.10
	10:00 AM	100	30.10
Midpoint: $P_0 = \$30.05$			Buy Spread Cost = \$0.05
			Sell Spread Cost = \$0.05
Buy Orders (Bids)	10:00 AM	100	30.00
	10:05 AM	500	30.00
	10:15 AM	200	30.00
	10:00 AM	500	29.90
	10:20 AM	100	29.90
	10:10 AM	500	29.80
	10:15 AM	1000	29.80

Input Variables & Calculations



Trade Quantity:

$$Size = \frac{Shares}{ADV} \quad (\%Adv, decimal)$$

Percentage of Volume:

$$POV = \frac{\frac{Actual/Historical/Past}{Shares}}{Actual Volume over Trade Period} \quad (decimal)$$

$$POV = \frac{\frac{Forecasted/Predicted/Future}{Size}}{Size + Time} \quad (decimal)$$

Trade Time:

$$Time = Size \cdot \frac{(1 - POV)}{POV}$$

Definitions:

Size = Order Size expressed as pct of ADV

ADV = Average Daily Volume (Moving Average)

POV = Percentage of Market Volume

Trade Time = Volume Time

σ = Annualized Volatility (Moving Avg, decimal)

Side = Order Side, +1 = Buy, -1 = Sell

Input Variables & Calculations



Average Daily Volume:

$$ADV(t) = \frac{1}{n} \cdot \sum_{i=1}^n Volume(t-i)$$

Median Daily Volume:

$$MDV(t) = Median(Volume(t-1), \dots, Volume(t-n))$$

Annualized Volatility

$$r_t = \ln\left(\frac{P_t}{P_{t-1}}\right)$$

$$\sigma(t) = \sqrt{250 \cdot \frac{1}{n-1} \cdot \sum_{t=1}^n (r_t - \bar{r})^2}$$

Forecasted Period Volume:

$$\hat{V}(t) = ADV \cdot f(v(t)) \cdot Adj$$

Definitions

ADV = Average Daily Volume (Moving Average)

MDV = MedianDaily Volume (Moving Average)

Volume(t-i) = Actual Total Volume on day t-i

$\hat{V}(t)$ = Predicted Volume in Time Period

P_t = Price on day t

r_t = Log Return on day t

σ = Annualized Volatility (Moving Average as decimal)

n = historical period, commonly 30 days

$f(v(t))$ = percentage of historical daily volume in period t

Adj = volume adjustment due to Day of Week, Month or Quarter End, Earnings, Fed Day, Special Event.

Market Observations

What MI Explanatory Factors are Important?



- Share Quantity
 - Size (%Adv)
- Volatility
 - Annualized
- POV / Trading Rate

Developing Market Impact Model

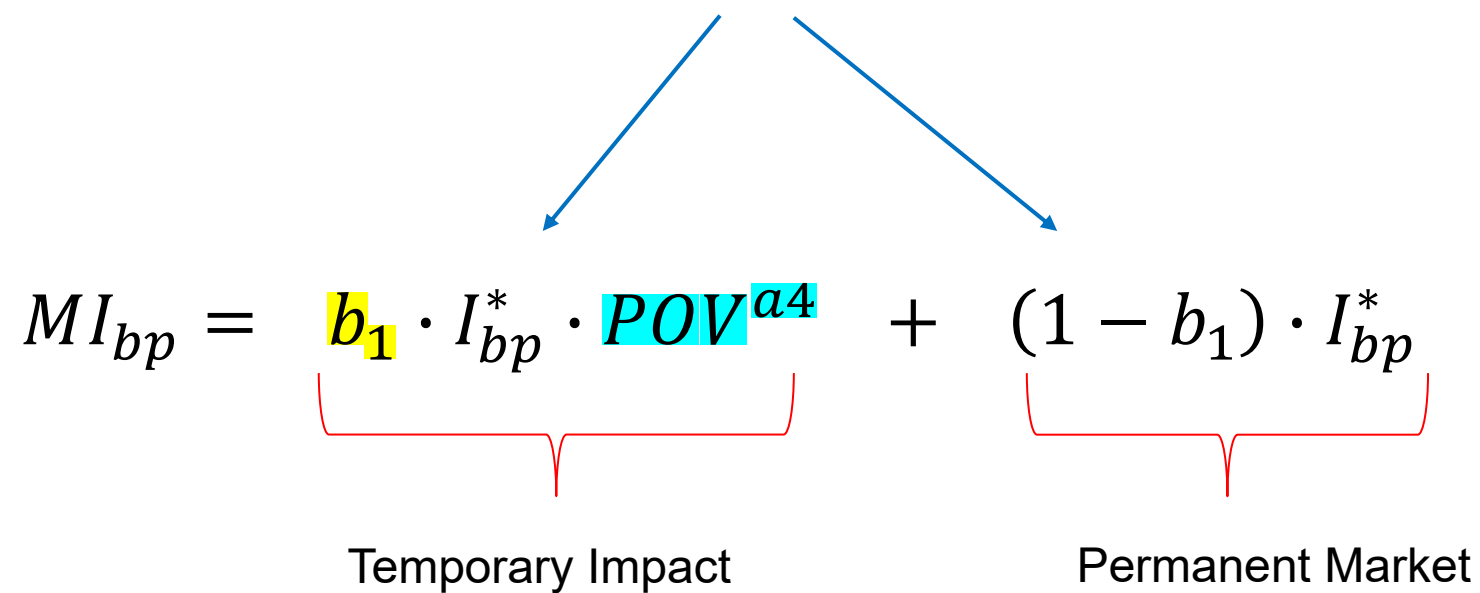
I-Star Model



$$I_{bp}^* = a_1 \cdot Size^{a_2} \cdot \sigma^{a_3}$$

I-Star Model

$$I_{bp}^* = a_1 \cdot Size^{a_2} \cdot \sigma^{a_3}$$


$$MI_{bp} = \underbrace{b_1 \cdot I_{bp}^* \cdot POV^{a_4}}_{\text{Temporary Impact}} + \underbrace{(1 - b_1) \cdot I_{bp}^*}_{\text{Permanent Market}}$$

I-Star Market Impact Model



$$I_{bp}^* = a_1 \cdot Size^{a_2} \cdot \sigma^{a_3}$$

$$MI_{bp} = b_1 \cdot I_{bp}^* \cdot POV^{a_4} + (1 - b_1) \cdot I_{bp}^*$$

I-Star Market Impact Model



$$I_{bp}^* = a_1 \cdot Size^{a_2} \cdot \sigma^{a_3}$$

$$MI_{bp} = b_1 \cdot I_{bp}^* \cdot POV^{a_4} + (1 - b_1) \cdot I_{bp}^*$$

$$MI_{bp} = I_{bp}^* \cdot (b_1 \cdot POV^{a_4} + (1 - b_1))$$

I-Star Market Impact Model



$$I_{bp}^* = a_1 \cdot Size^{a_2} \cdot \sigma^{a_3}$$

$$MI_{bp} = b_1 \cdot I_{bp}^* \cdot POV^{a_4} + (1 - b_1) \cdot I_{bp}^*$$

$$MI_{bp} = I_{bp}^* \cdot (b_1 \cdot POV^{a_4} + (1 - b_1))$$

$$MI_{bp} = (a_1 \cdot Size^{a_2} \cdot \sigma^{a_3}) \cdot (b_1 \cdot POV^{a_4} + (1 - b_1))$$

Estimating MI Parameters: Non-Linear Regression



I-Star Model:

$$MI = I^* \cdot (b_1 \cdot POV^{a_4} + (1 - b_1))$$

$$MI = (a_1 \cdot Size^{a_2} \cdot \sigma^{a_3}) \cdot (b_1 \cdot POV^{a_4} + (1 - b_1))$$

Non-Linear Regression:

$$\text{Min } f = \sum (\text{Arrival Cost} - MI)^2$$

$$\text{Min } f = \sum \left(\text{Actual Cost} - \left((a_1 \cdot Size^{a_2} \cdot \sigma^{a_3}) \cdot (b_1 \cdot POV^{a_4} + (1 - b_1)) \right) \right)^2$$

Subject to:

$$0 \leq b_1 \leq 1$$

$$a_1, a_2, a_3, a_4 \geq 0$$

where,

Cost = Measured Trade Cost (difference between execution price and arrival price)

Example: Python #1



Estimate Market Impact Parameters using Python

- Python Script: [KRG_Est MI_Parameters.ipynb](#)
- File: [MIData_KRG.csv](#)

Estimating MI Parameters: Results

MI Parameters	
Parameter	Value
a1	883.25
a2	0.35
a3	0.76
a4	0.83
b1	0.96

I-Star Model:

$$I^* = a_1 \cdot Size^{a_2} \cdot \sigma^{a_3}$$

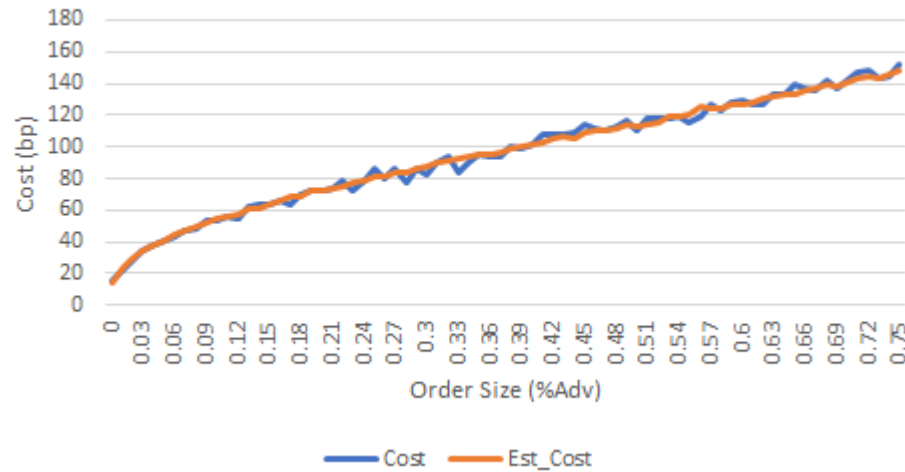
$$MI = I^* \cdot (b_1 \cdot POV^{a_4} + (1 - b_1))$$

$$I^* = 883.25 \cdot Size^{0.35} \cdot \sigma^{0.76}$$

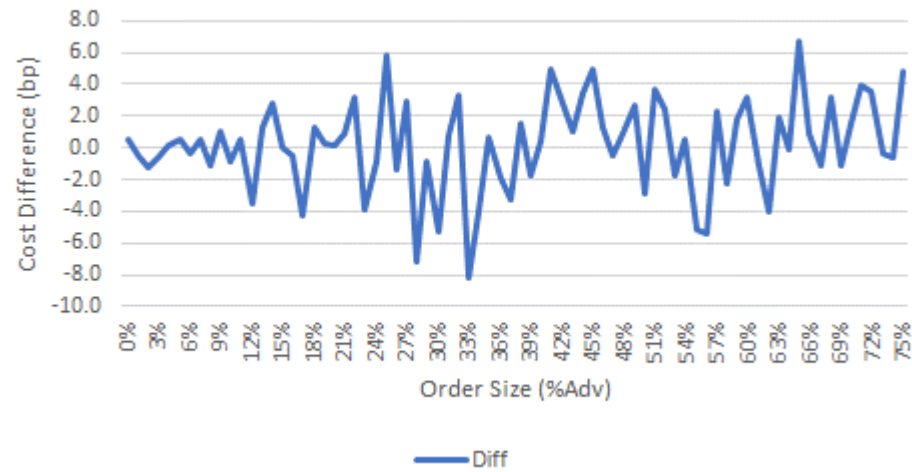
$$MI = I^* \cdot (0.96 \cdot POV^{0.83} + (1 - 0.96))$$

Comparison of Actual to Estimated

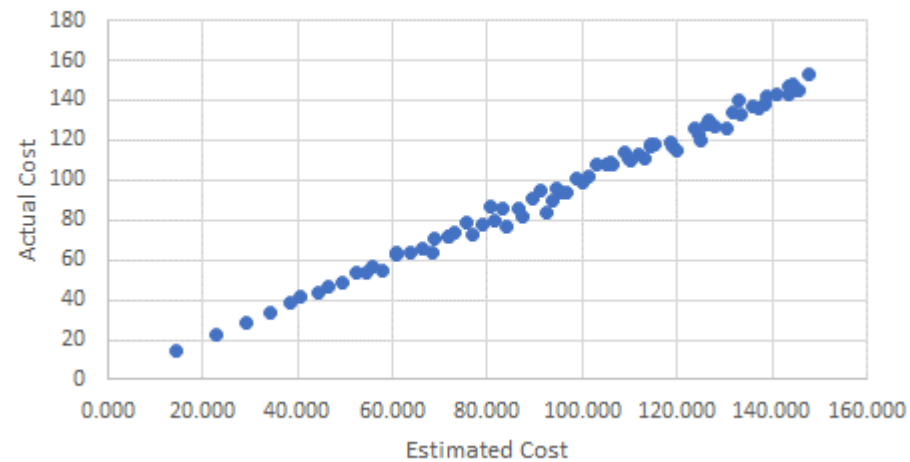
Comparison of Actual to Estimated



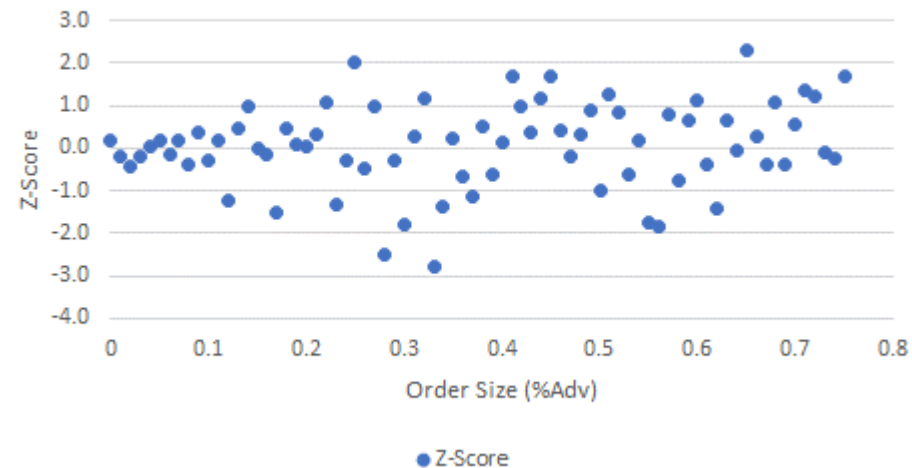
Comparison of Actual to Estimated



Cost Comparison

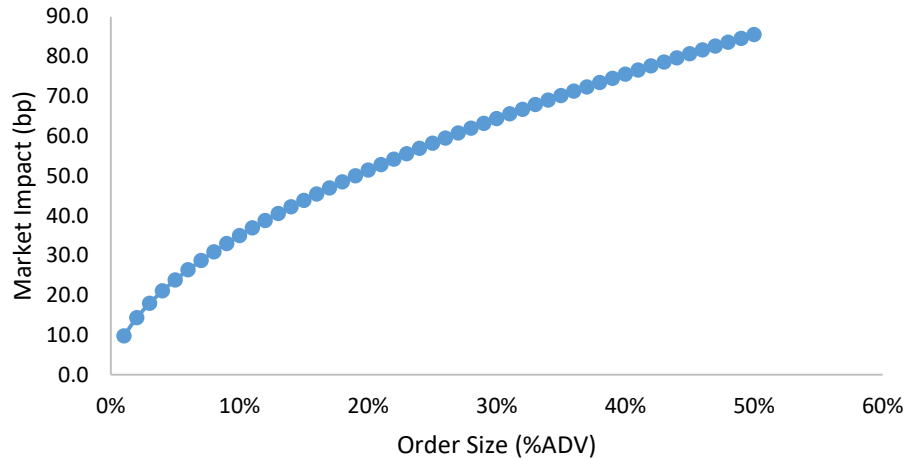


Z-Score

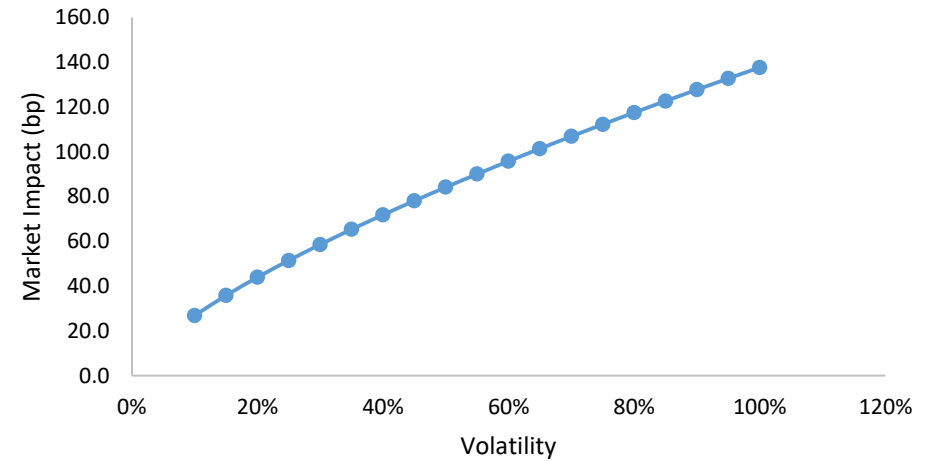


Market Impact Relationships

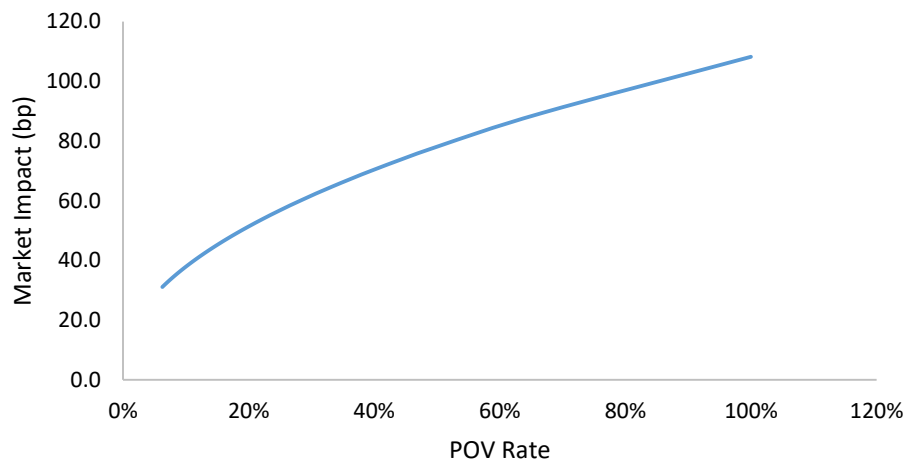
Market Impact Cost as function of Size



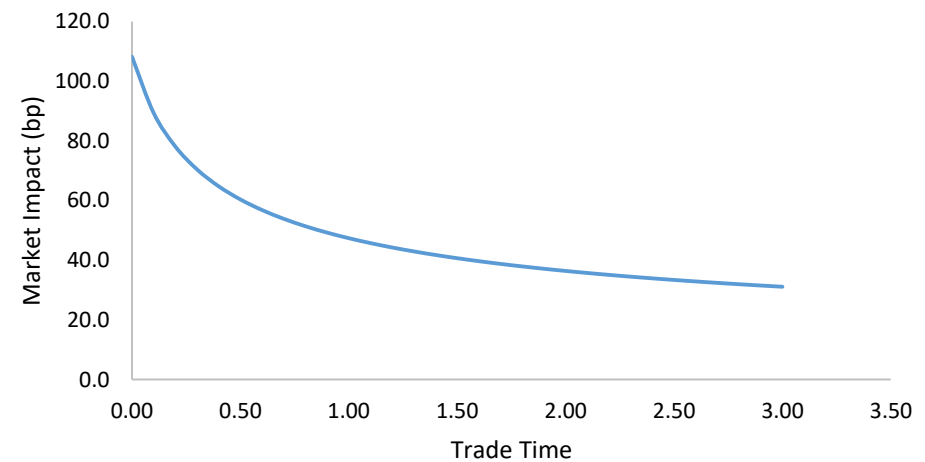
Market Impact as function of Volatility



Market Impact Cost as function of POV Rate



Market Impact Cost as function of Time



Trading Analytics

Pre-Trade

- Cost Curves
- Basket Analysis

Intra-Day

- Real-Time

Post-Trade

- Cost Curves
- Basket Analysis

Demonstrate how to develop each analytic in Excel using KRG Analytics Suite

Pre-Trade Analysis



Pre-Trade Analysis: Pre-trade analysis provides investors with the estimated trading costs of an order. This includes the expected market impact, price appreciation, and timing risk. Traders also use pre-trade analysis to evaluate different trading strategies and algorithms based on trading cost and risk. They also allow investors to incorporate their own market views and proprietary alpha forecasts directly into the analyses so that investors can develop a customized analysis for their specified order. Traders use these functions to perform single stock and portfolio multi-period trade schedule optimization.

- The goal of pre-trade TCA is to provide traders with the necessary information to select the most appropriate algorithm and strategy based on the underlying investment objectives of the fund.

Pre-Trade Analysis

Pre-Trade Model

MI Parameters	
a1=	746.1864
a2=	0.3521
a3=	0.7776
a4=	0.9100
b1=	0.9500

Strategy	POV	Time	MI	TR
POV	10%	2.1	17.0	80.6
Time	16%	1.0	27.2	61.8
Optimal	11%	1.6	18.8	73.4

Num	Symbol	Side	Shares	Weight	Size	POV Strategy				VWAP Strategy				Single Stock Optimization			
						POV	Time	MI	TR	POV	Time	MI	TR	POV	Time	MI	TR
1	A	Buy	415,479	3.46%	15.00%	10%	1.35	17.3	79.0	13.0%	1.0	20.60	79.04	11.5%	1.15	18.93	73.09
2	AAL	Sell	111,250	0.11%	0.47%	10%	0.04	6.8	20.3	0.5%	1.0	2.34	20.30	8.5%	0.05	6.16	22.20
3	AAP	Buy	125,666	1.59%	1.06%	10%	0.10	6.7	20.8	1.0%	1.0	2.63	20.77	8.5%	0.11	6.09	22.72
4	AAPL	Buy	47,676	0.46%	4.07%	10%	0.37	13.2	52.7	3.9%	1.0	7.91	52.69	10.5%	0.35	13.65	51.28
5	ABBV	Buy	497,060	3.59%	1.06%	10%	0.10	5.8	17.1	1.1%	1.0	2.25	17.05	8.5%	0.11	5.22	18.65
6	ABC	Buy	293,879	2.35%	3.56%	10%	0.32	10.0	36.4	3.4%	1.0	5.64	36.44	9.5%	0.34	9.66	37.49
7	ABMD	Buy	36,839	0.84%	1.85%	10%	0.17	7.5	24.3	1.8%	1.0	3.35	24.33	9.0%	0.19	6.99	25.79
8	ABT	Buy	430,812	3.87%	8.10%	10%	0.73	13.5	55.9	7.5%	1.0	11.33	55.94	10.5%	0.69	13.94	54.44
9	ACN	Buy	182,047	3.40%	5.68%	10%	0.51	13.3	54.0	5.4%	1.0	9.29	53.96	10.5%	0.48	13.74	52.52
10	ADBE	Buy	7,914	0.32%	0.32%	10%	0.03	5.4	14.8	0.3%	1.0	1.78	14.79	8.0%	0.04	4.71	16.71
11	ADI	Buy	6,362	0.06%	0.19%	10%	0.02	4.6	11.7	0.2%	1.0	1.47	11.75	7.5%	0.02	3.86	13.75
12	ADM	Buy	1,072,359	4.12%	45.30%	10%	4.08	25.2	135.2	31.2%	1.0	57.20	135.16	13.0%	3.03	29.95	116.55
13	ADP	Buy	132,582	1.53%	3.69%	10%	0.33	12.4	48.3	3.6%	1.0	7.11	48.26	10.0%	0.33	12.40	48.26
14	ADSK	Sell	6,105	0.12%	0.32%	10%	0.03	5.7	15.7	0.3%	1.0	1.87	15.67	8.0%	0.04	4.93	17.72
15	AEE	Buy	571,123	3.73%	113.17%	10%	10.18	31.0	184.0	53.1%	1.0	108.33	183.99	14.0%	6.95	38.73	152.00
16	AEP	Buy	812,402	5.48%	33.48%	10%	3.01	16.5	77.3	25.1%	1.0	31.63	77.31	11.5%	2.58	18.07	71.49
17	AES	Buy	404,373	0.60%	17.02%	10%	1.53	20.6	99.6	14.5%	1.0	26.45	99.58	12.0%	1.25	23.19	89.89
18	AFL	Buy	216,708	0.65%	16.32%	10%	1.47	20.7	100.2	14.0%	1.0	25.97	100.24	12.0%	1.20	23.35	90.48
19	AIG	Sell	323,408	0.74%	12.13%	10%	1.09	22.5	109.6	10.8%	1.0	23.64	109.63	12.0%	0.89	25.31	98.96
20	AIV	Buy	329,337	0.92%	6.26%	10%	0.56	13.9	57.3	5.9%	1.0	10.18	57.32	10.5%	0.53	14.35	55.79

Intra-Day Analysis: Intra-day analysis provides investors with the necessary real-time analytics to monitor transaction costs during trading. These models provide investors with point in time trading costs estimates (for executed shares) and the projected trading costs that will result from completing the order (for those shares that still need to be executed). This is accomplished by incorporating market momentum and actual market conditions (volume, volatility, and aggregated imbalances) directly into the analysis.

- The goal of Intra-Day analysis is to provide investors with the necessary information to determine when it is advantageous to take advantage of changing market conditions and favorable opportunities.

Cost Curve



Cost Curve: A cost curve provides the Portfolio Manager with insight into the cost of executing a specified number of shares utilizing a specified trading strategy commonly expressed in terms of percentage of volume (POV) or trade time (Time).

- The goal of cost curves is to provide investors with the necessary information to determine how many shares can be absorbed into the market at the managers level of price sensitivity.

Cost Curves

Cost Curves												
Order												
Symbol:	A											
Volatility:	0.186											
ADV:	2,770,507											
Price:	100.94											
Order Statistics			Market Impact Estimates - basis points									
Size			POV Strategy									
(% Adv)	Shares	Dollars	5%	10%	15%	20%	25%	30%	35%	40%	45%	50%
1%	27,705	2,796,550	4.5	6.7	8.7	10.8	12.7	14.7	16.6	18.5	20.3	22.2
5%	138,525	13,982,749	7.9	11.7	15.4	19.0	22.5	25.9	29.2	32.6	35.8	39.1
10%	277,051	27,965,498	10.1	15.0	19.7	24.2	28.7	33.0	37.3	41.6	45.7	49.9
15%	415,576	41,948,246	11.6	17.3	22.7	27.9	33.1	38.1	43.0	47.9	52.8	57.6
20%	554,101	55,930,995	12.9	19.1	25.1	30.9	36.6	42.1	47.6	53.0	58.4	63.7
25%	692,627	69,913,744	13.9	20.7	27.2	33.4	39.6	45.6	51.5	57.4	63.2	68.9
30%	831,152	83,896,493	14.8	22.1	29.0	35.7	42.2	48.6	54.9	61.2	67.4	73.5
35%	969,677	97,879,242	15.7	23.3	30.6	37.6	44.5	51.3	58.0	64.6	71.1	77.6
40%	1,108,203	111,861,991	16.4	24.4	32.0	39.5	46.7	53.8	60.8	67.7	74.5	81.3
45%	1,246,728	125,844,739	17.1	25.5	33.4	41.1	48.7	56.1	63.4	70.6	77.7	84.7
50%	1,385,254	139,827,488	17.8	26.4	34.7	42.7	50.5	58.2	65.8	73.2	80.6	87.9
55%	1,523,779	153,810,237	18.4	27.3	35.9	44.1	52.2	60.2	68.0	75.7	83.4	90.9
60%	1,662,304	167,792,986	18.9	28.2	37.0	45.5	53.8	62.0	70.1	78.1	86.0	93.8
65%	1,800,830	181,775,735	19.5	29.0	38.0	46.8	55.4	63.8	72.1	80.3	88.4	96.4
70%	1,939,355	195,758,484	20.0	29.7	39.0	48.0	56.9	65.5	74.0	82.4	90.8	99.0
75%	2,077,880	209,741,232	20.5	30.5	40.0	49.2	58.3	67.1	75.8	84.5	93.0	101.4
80%	2,216,406	223,723,981	21.0	31.2	40.9	50.4	59.6	68.7	77.6	86.4	95.1	103.8
85%	2,354,931	237,706,730	21.4	31.8	41.8	51.4	60.9	70.1	79.3	88.3	97.2	106.0
90%	2,493,456	251,689,479	21.8	32.5	42.6	52.5	62.1	71.6	80.9	90.1	99.2	108.2
95%	2,631,982	265,672,228	22.3	33.1	43.5	53.5	63.3	72.9	82.4	91.8	101.1	110.2
100%	2,770,507	279,654,977	22.7	33.7	44.3	54.5	64.5	74.3	83.9	93.5	102.9	112.2

Post-Trade Analysis: Post-trade analysis serves as a report card on the trade. It provides investors with the cost of the trade and an evaluation of the trading performance. Post-trade analysis will include a cost comparison to various price benchmarks (e.g., Arrival, Open, VWAP, Close, T-1, and T+1) as will also compare actual costs to a pre-trade trading cost estimate for the selected strategy.

Post-trade analysis allows funds to determine how well their brokers and algorithms performed given actual market conditions. Investors can rank their brokers and their algorithms to determine which brokers and algorithms are adding value to the fund and determine which algorithms and which brokers may be underperforming expectations and causing the fund to incur unnecessary higher trading costs. Investors can compute additional statistics related to the trade such as the relative performance measure (RPM) to help determine which brokers are adding value to the trading process and which brokers are causing funds to incur unnecessary trading costs. Customized post trade reports provide clients with the ability to sort, filter, and evaluate different trading situations right on their own desktop.

- The goal of Post-Trade analysis is to help investors measure trading costs and evaluate trade performance.
- It also helps investors determine which broker and which broker algorithms provide the best results based on order characteristics and market conditions.

Post-Trade Analysis

Post-Trade Analysis (TCA)

				TCA: Benchmark Cost Analysis						TCA: Implementation Shortfall					
				Open	Arrival	VWAP	Close	Value-		Delay	Exec.	Opportunity	Commission		
				Cost (bp)	Cost (bp)	Cost (bp)	Cost (bp)	Add (bp)	Z-Score	Cost (\$)	Cost (\$)	Cost (\$)	Cost (\$)	Total IS (\$)	Total IS (bp)
Broker															
			Broker A	14.3	14.3	42.0	-20.3	4.4	0.09	0	-69,238	21,157	16,817	-31,264	-1.39
			Broker B	-30.2	-30.2	25.1	37.9	-4.7	-1.54	0	-135,751	0	16,303	-119,449	-14.98
			Broker C	-10.0	-10.0	28.4	18.4	8.3	0.42	0	-309,788	164,834	41,577	-103,377	-2.94
			Broker D	-0.3	-0.3	-9.7	-31.4	-7.0	0.13	0	140,425	24,057	22,181	186,662	6.43
			Broker E	-18.2	-18.2	12.3	73.0	11.5	2.83	0	-493,213	398	28,164	-464,651	-17.53

Order Details				TCA: Benchmark Cost Analysis						TCA: Implementation Shortfall					
Num	Symbol	Side	Broker	Open	Arrival	VWAP	Close	Value-		Delay	Exec.	Opportunity	Commission		
				Cost (bp)	Cost (bp)	Cost (bp)	Cost (bp)	Add (bp)	Z-Score	Cost (\$)	Cost (\$)	Cost (\$)	Cost (\$)	Total IS (\$)	Total IS (bp)
1	A	Buy	Broker C	-23.8	-23.8	92.2	78.1	41.1	0.52	0	-99,715	0	4,155	-95,560	-22.8
2	AAL	Sell	Broker D	-8.1	-8.1	48.5	-224.4	14.4	0.80	0	890	-5,785	890	-4,005	-29.3
3	AAP	Buy	Broker C	-28.0	-28.0	42.6	386.8	34.8	1.67	0	-54,036	0	1,257	-52,780	-27.4
4	AAPL	Buy	Broker B	-1.7	-1.7	26.8	-213.0	15.0	0.28	0	-954	0	477	-477	-0.9
5	ABBV	Buy	Broker E	-59.4	-59.4	-20.6	61.2	65.1	3.82	0	-258,471	0	4,971	-253,500	-58.2
6	ABC	Buy	Broker E	-29.9	-29.9	12.4	150.2	39.9	1.10	0	-85,225	0	2,939	-82,286	-28.9
7	ABMD	Buy	Broker E	52.3	52.3	68.7	42.5	-45.0	-1.89	0	51,280	398	354	52,032	51.0
8	ABT	Buy	Broker A	6.4	6.4	48.9	149.1	6.9	0.13	0	29,252	-19,774	4,179	13,657	2.9
9	ACN	Buy	Broker C	-36.7	-36.7	-106.3	96.4	50.0	0.93	0	-151,099	0	1,820	-149,279	-36.3
10	ADBE	Buy	Broker A	-5.3	-5.3	-36.4	111.8	10.7	0.73	0	-2,037	-450	78	-2,408	-6.2
11	ADI	Buy	Broker A	-16.3	-16.3	-75.8	-112.0	20.9	1.78	0	-1,209	0	64	-1,145	-15.4
12	ADM	Buy	Broker C	-38.7	-38.7	-89.9	355.5	63.9	0.47	0	-193,025	0	10,724	-182,301	-36.6
13	ADP	Buy	Broker D	39.4	39.4	117.0	-58.2	-27.0	-0.56	0	72,920	0	1,326	74,246	40.1
14	ADSK	Sell	Broker D	0.9	0.9	44.8	289.2	4.8	0.31	0	-122	0	61	-61	-0.4
15	AEE	Buy	Broker D	-19.0	-19.0	-157.1	232.0	49.9	0.27	0	-85,668	0	5,711	-79,957	-17.7

Trade Cost Equations

Forecasting Equations - Data

	<u>MI</u>	<u>TR</u>	<u>PA</u>
Parameters	yes	no	no
Size	yes	yes	yes
Volatility	yes	yes	no
AlphaBp	no	no	yes
Side	no	no	yes
POV	yes	yes	yes
Trade Time	yes	yes	yes